

Review

Direct Interface Circuits for Resistive, Capacitive, and Inductive Sensors: A Review

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Abstract: Direct interface circuits (DICs) connect resistive, capacitive, and inductive sensors directly to microcontrollers or FPGAs, eliminating analog conditioning stages and offering compact, low-cost, and low-power instrumentation. This systematic review qualitatively synthesizes research up to March 2025 on DIC operation principles, performance metrics, and application domains. Following PRISMA guidelines, 90 studies from IEEE Xplore, ScienceDirect, MDPI, SpringerLink, Scopus, and Google Scholar were selected based on predefined inclusion criteria. Most studies focused on *RC*-based circuits (53%), followed by *RL*-based (5%) and charge transfer capacitive interfaces (5%). *RC*-DICs demonstrated accuracies below 0.01% using adaptive calibration; *RL*-DICs achieved resolutions of 10–12 bits with higher cycle requirements, while charge transfer interfaces presented systematic errors up to $\pm 5\%$ due to parasitic capacitances. Environmental monitoring, biomedical sensing, liquid-level control, and vehicular detection were frequent application fields. Due to methodological heterogeneity, findings were synthesized qualitatively without quantitative meta-analysis or formal bias assessments. Future research directions include enhanced noise immunity, simplified calibration, and robust parasitic effect compensation.



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Keywords: direct interface circuits (DICs); field-programmable gate arrays (FPGAs); microcontrollers (MCUs); low-power instrumentation; sensor interface; review

1. Introduction

The rapid advancement of technology continues to pose new challenges in electronic instrumentation. There is an increasing demand for innovative, compact, and energy-efficient electronic systems capable of monitoring wireless sensors and sensor interfaces in various domains, including the Internet of Things (IoT) [1], wearable smart sensors for vital sign monitoring [2], and sensors deployed in smart city infrastructures [3]. These areas have garnered significant attention from the scientific community due to their broad impact and wide range of potential applications.

In many cases, the measurement of electrical quantities such as resistance, capacitance, and inductance is essential for interpreting sensor responses associated with physical phenomena (e.g., pressure, distance, temperature, etc.). Figure 1a shows a traditional sensor interface architecture, wherein the sensor signal is processed through an analog signal conditioning stage that typically involves amplification, filtering, linearization, and analog-

to-digital conversion (ADC) [4], followed by digital processing using a microcontroller unit (MCU) or Field-Programmable Gate Array (FPGA) [5].

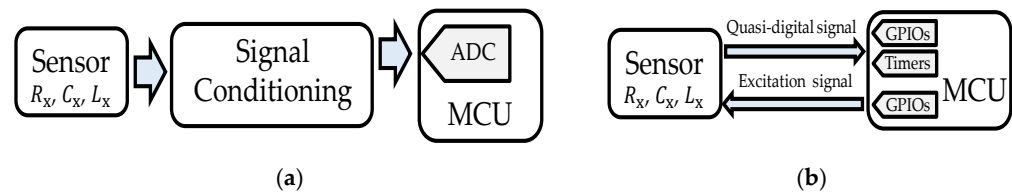


Figure 1. (a) Traditional sensor interface; (b) direct sensor interface approach.

In recent years, an alternative approach to electronic instrumentation has emerged for measuring sensors with resistive, capacitive, or inductive variations without the need for traditional signal conditioning stages. This technology, known as direct sensor–microcontroller interface circuits, or direct interface circuits (DICs), has gained acceptance because of its simplicity and efficiency [6,7]. Initially introduced in application notes by semiconductor manufacturers, DICs have proven to be highly practical for sensor interfacing.

Early implementations are generally categorized into charge time-based [8–11] and discharge time-based circuits [12,13]. Both types utilize RC configurations, where a digital processor excites the analog sensor to produce a time-domain modulated signal. This signal is digitized via a time-to-digital conversion (TDC) using the processor’s internal timer and GPIOs, as shown in Figure 1b. The resulting signals, commonly referred to as quasi-digital signals [14], have formed the basis for numerous studies targeting resistive [15–17], capacitive [18–20], and inductive [21,22] sensors. These approaches enable compact, energy-efficient sensor interfaces with low active-mode power consumption, reducing overall power draw and eliminating the need for extra components. As a result, they minimize circuit size and cost [7].

DICs could emerge in areas where real-time data acquisition is necessary, which facilitates quick and direct measurements from sensors (e.g., health monitoring, environmental sensing, and smart home systems), and its simplicity in scalability for IoT systems where more of one sensor can be added without significantly increasing the complexity of the circuit, leading an energy efficient system, making it feasible to create large networks of interconnected devices. However, challenges remain in achieving high accuracy, resolution, and optimal measurement time, as these performance metrics are closely influenced by uncertainties in the time-to-digital conversion process [23]. Additionally, these circuits are susceptible to interference from power supply fluctuations [24] and uncertainties in detecting quasi-digital signals. These uncertainties are primarily related to the voltage thresholds V_{TL} and V_{TH} used to detect the sensor signal at a specific voltage level V_c of the digital processor, directly impacting the resolution and reliability of the measurement process [6].

1.1. Related Works

This review serves as a follow-up to the works presented by Reverter in [6,7], which discuss the art of directly interfacing sensors to microcontrollers (MCUs), building upon application notes from semiconductor manufacturers [8–13]. Reverter established fundamental design guidelines to simplify sensor interface circuitry, focusing on power efficiency and reliable performance. It highlighted the cost-effectiveness of such approaches for various resistive and capacitive sensors and proposed future directions for broadening their applicability in modern electronic systems. Additionally, Reverter in [25] presented a book chapter discussing DIC’s principles and applications for some commercial sensors, and [26] related to advanced techniques for resistive sensors and their performance in some applications.

However, the scope of these early works was primarily centered on explaining the operating principles of direct interface techniques and evaluating their performance, efficiency metrics, and applications in commercial sensors at that time. These insights motivated a deeper investigation into this research area to explore recent advancements and emerging trends. The aim of this paper is to provide a literature review with a qualitative synthesis of the state of the art of published papers on DICs up to the year 2025. This includes a detailed examination of their key components and the application of advanced techniques to enhance accuracy, resolution, and power efficiency. The contributions of this review are as follows:

- An overview of the operating principles, core components, and implementation techniques of DICs for resistive, capacitive, and inductive sensors.
- A critical analysis of recent publications proposing improved methods to enhance measurement accuracy, resolution, acquisition time, and uncertainty management in DIC-based systems.
- An assessment of current application domains for DICs and strategic recommendations for future research efforts.
- A discussion of the most significant advances and remaining technical challenges that must be addressed for broader adoption of DICs in future systems.

1.2. Paper Organization

The structure of this paper is outlined in Figure 2. Section 1 introduces DICs as an alternative for sensor interfacing and presents the main contributions and review methodology. Section 2 offers an overview of the key elements and characteristics of DIC implementation. Section 3 details various types of direct interface circuits, explaining their operating principles and reviewing enhanced techniques based on RC , RL , and capacitive charge transfer methods. Section 4 analyzes the literature results by circuit type, sensor type (resistive, capacitive, and inductive), and digital processors used in DICs, concluding with relevant application areas. Section 5 discusses review findings, emphasizing recent advancements, challenges, and strategic recommendations for future research. Finally, Section 6 summarizes the conclusions of the study.

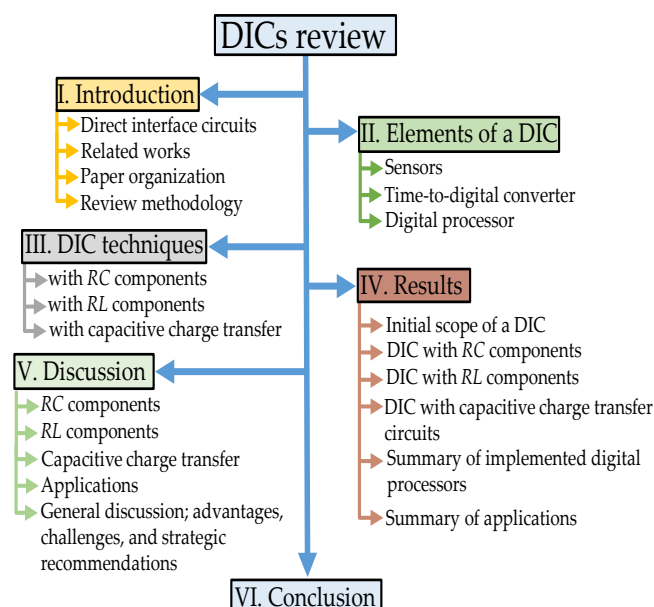


Figure 2. Paper organization.

1.3. Methodology of Review

This work was developed following the PRISMA statement guidelines for systematic literature reviews [27], which provide a structured approach for presenting results and ensuring transparency, thereby enhancing the quality of the review. Figure 3 outlines our search strategy, which involved identifying relevant articles using specific keywords and inclusion criteria across several well-established digital databases, including IEEE Xplore, ScienceDirect, MDPI, and SpringerLink, as well as bibliographic search engines such as Google Scholar and Scopus.

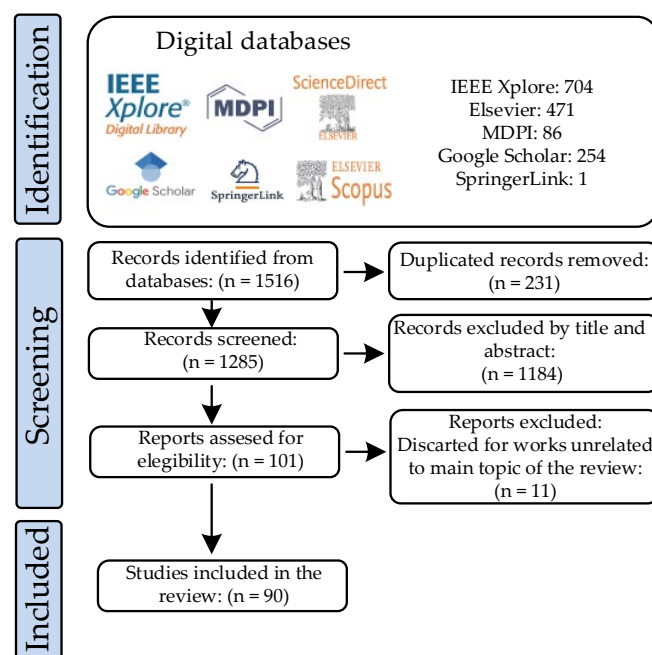


Figure 3. The PRISMA method scheme for this review.

The primary inclusion keywords used were “direct interface circuits” and “sensor interfaces”. An initial screening process employed publication titles and abstracts, guided by exclusion and inclusion criteria, to filter out unrelated content.

The selected documents included peer-reviewed journal articles, application notes from semiconductor manufacturers, and technical books related to sensor interfacing. The inclusion criteria for the literature review were as follows: (i) Publications describing sensor interfaces based on the operating principles of direct connection to a digital processor (e.g., MCU or FPGA); (ii) studies presenting novel methods or enhanced techniques related to DICs; (iii) analytical reviews or comparative studies; (iv) application notes demonstrating practical implementations of DICs.

This selection process enabled the identification of relevant full-text articles with critical insights into the operational principles, measurement ranges, accuracy, resolution, acquisition time, sources of systematic error, and power consumption associated with DIC-based systems. This work selected a total of 106 references—including papers, books, book chapters, application notes, and support papers—for this review, focusing on RC , RL , and capacitive charge transfer configurations.

Figure 4 provides a classified overview of the selected works, highlighting the distribution of the reviewed literature: DICs with RC components represent the most extensively studied field, accounting for 53% of the review, followed by DICs with RL components (5%), DICs employing capacitive charge transfer techniques (5%), general application notes reported in the literature (22%), and support papers (16%).

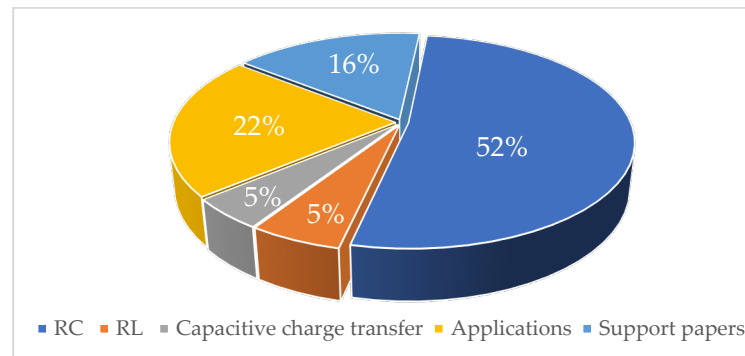


Figure 4. Classification of reviewed papers.

Excluding the supporting papers that introduce relevant concepts and related works, Figure 5 presents a general graph of the included works in this review for analysis and discussion ($n = 90$), organized by year and publisher.

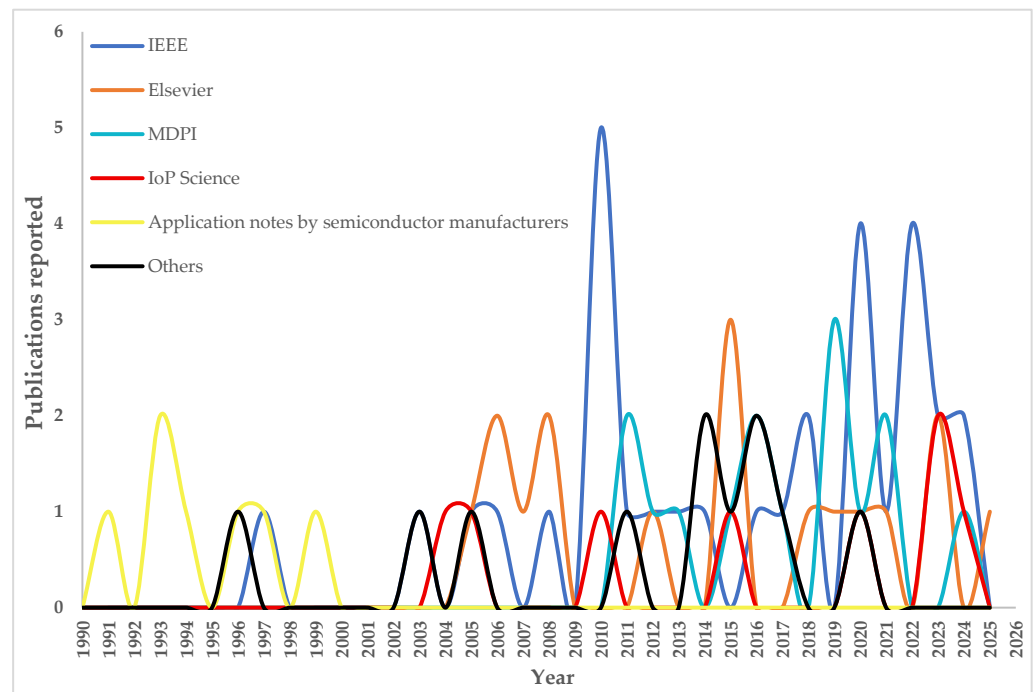


Figure 5. A publication scheme of the articles included in the review ($n = 90$) by year.

2. Elements of a DIC

The fundamental elements of a DIC connecting a sensor to a digital processor are designed to minimize the use of external components. These elements facilitate the excitation of the sensor and the acquisition of a quasi-digital signal in the time domain, which is subsequently digitized by a time-to-digital converter (TDC). This process enables the estimation of the corresponding electrical variable—resistance, capacitance, or inductance—without the need for traditional analog signal conditioning, as shown in Figure 6. Within this context, the following sections examine a summary of the key components and their characteristics to provide a clear understanding of the operating principles underlying DIC implementations.

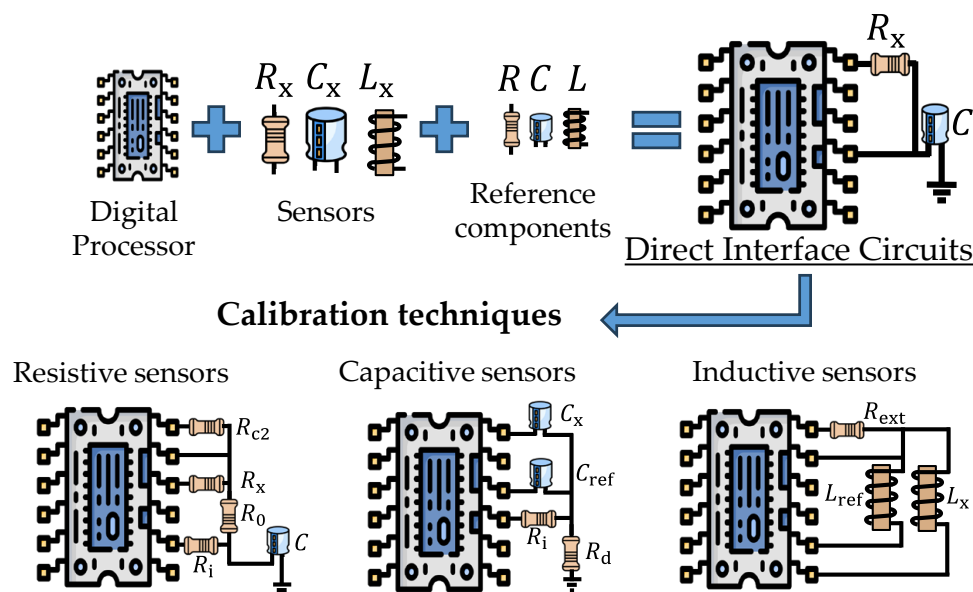


Figure 6. Scheme of a basic DIC and its calibration techniques for resistive, capacitive, and inductive sensors.

2.1. Sensors

Pallás-Areny et al. [28] describe resistive sensors as devices that change their electrical resistance R , either directly or indirectly, in response to variations in a physical quantity. These sensors can be categorized into three main types, as illustrated in Figure 7. The first type, simple resistive sensors, consists of a single resistive element R_x , providing a straightforward and effective solution for basic sensing applications. The second type, differential resistive sensors, includes two sensing elements, R_{x1} and R_{x2} , sharing a common terminal and showing opposite response variations. This arrangement enhances accuracy by minimizing noise and improving measurement reliability in high-precision environments. The third type, bridge-type resistive sensors, utilizes one, two, or four sensing elements configured in quarter-bridge, half-bridge, or full-bridge Wheatstone arrangements. These resistive sensors are commonly used to measure temperature (e.g., platinum resistance thermometers and thermistors), light (e.g., Light-Dependent Resistors), gas (e.g., tin dioxide sensors), humidity, and displacement utilizing linear or rotary potentiometers.

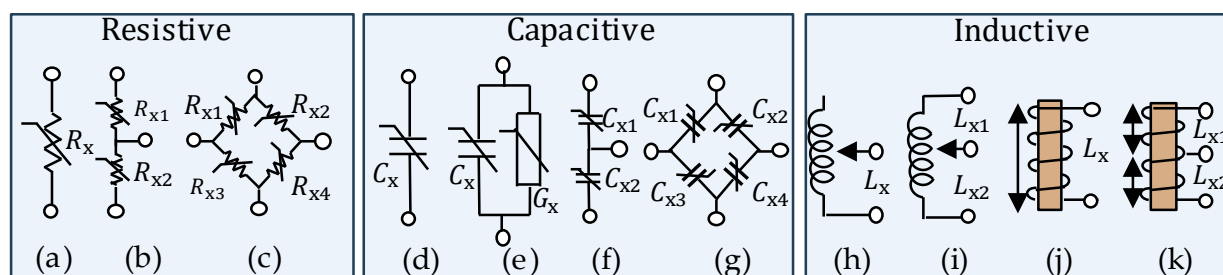


Figure 7. Resistive sensors: (a) simple, (b) differential, and (c) bridge; capacitive sensor types: (d) simple, (e) lossy, (f) differential, and (g) bridge; and inductive sensor types: (h) simple, (i) differential, (j) simple with ferromagnetic core, and (k) differential with ferromagnetic core.

Capacitive sensors, which represent another key category in sensor interfaces, exhibit changes in capacitance due to variations in dielectric properties or geometric factors such as plate area or distance. According to Pallás-Areny et al. [28], capacitive sensors can be categorized into four types, as depicted in Figure 7. Simple capacitive sensors utilize a single capacitance element C_x , providing an essential and effective sensing configuration.

Lossy capacitive sensors account for real-world effects by modeling parasitic conductance G_x in parallel with C_x , enhancing accuracy by considering energy losses. Differential capacitive sensors improve sensitivity and noise immunity by incorporating two elements, C_{x1} and C_{x2} , with a shared electrode and opposite response characteristics. Finally, bridge-type capacitive sensors employ one, two, or four sensing elements in bridge configurations to achieve high precision and stability. These sensors are utilized in various applications, including liquid level detection, humidity sensing, gas detection, pressure monitoring [7], and fluid concentration analysis [20].

Inductive sensors constitute the third major category commonly utilized in industrial applications to measure displacement between metallic objects and other physical quantities, such as pressure [29]. These sensors operate on variations in magnetic reluctance, which correspond to changes in magnetic flux driven by an electric current. When this current passes through the system, it is linked to inductance L_x . Pallás-Areny et al. [28] categorize inductive sensors into several types, four of which are illustrated in Figure 7. These include configurations with inductance variation based on the number of coil turns in both simple and differential setups, as well as designs that involve the movement of a ferromagnetic core, again in either simple or differential arrangements.

Reverter [30] presented a review of remote resistive sensors, which are situated in harsh, inaccessible environments or far from the main measurement and control systems. These remote sensors often involve parasitic resistances and temperature variations, which impact measurement accuracy. Some techniques for wire resistance are studied to compensate for the parasitic resistances of connecting leads. For instance, three-wire resistive sensors (Figure 8a) are designed to negate the parasitic resistances of the wires [31,32], with the third wire serving as the current supply, while four-wire resistive sensors (Figure 8b), known as the Kelvin connection, use four wires: two for providing the excitation current and two separate sense wires for measuring the voltage across the sensor [33]. A similar approach is presented by Hidalgo [34] and Nagarajan [35], who offered DICs for simple resistive sensors affected by lead-wire resistance. These methods employ techniques to reduce the resulting errors, as explained in detail in Section 4.

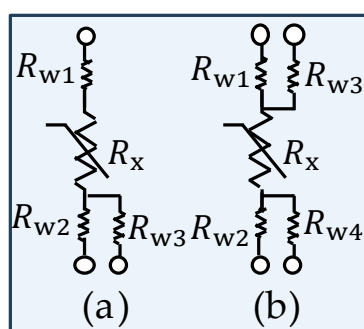


Figure 8. Wire-resistance techniques for resistive sensors: (a) three-wire resistive sensor, and (b) four-wire resistive sensor.

Resistive, capacitive, and inductive sensors are essential for converting physical magnitudes into measurable electrical signals. Given the focus of this work, a thorough understanding of sensor types and their configurations—whether simple, differential, or bridge-based—is crucial for developing efficient DIC techniques.

2.2. Time-to-Digital Conversion Principles and Uncertainty Sources

A time-to-digital converter (TDC) is a crucial electronic component in systems that require precise and accurate measurement of time intervals. Its primary function is to convert time-domain information, typically defined by the interval between two signal

events, such as pulse width or period, into a digital representation. A simplified explanation of TDC operation is provided by Henzler [36], who refers to it as a digital technique based on simple counting. In this approach, a time interval T_x is measured by counting the number of cycles of a reference clock T_{ref} using a universal counter, as illustrated in Figure 9. The interval is defined by the rising edges of a start and stop signal, and the result is digitized as T_s . These events are generally asynchronous to the reference clock T_{ref} in a universal counter. Timing errors may be introduced at both the start (T_{Start}) and stop (T_{Stop}) edges due to the limited resolution of T_{ref} . This resolution can be improved by increasing the clock frequency; however, this also leads to higher power consumption and imposes additional design constraints.

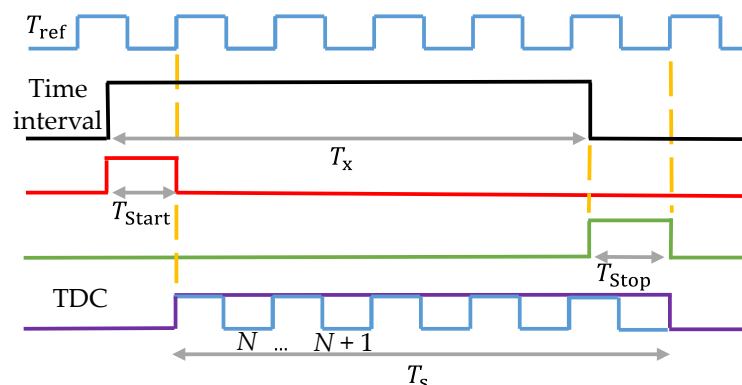


Figure 9. Time-to-digital converter principle.

Figure 10a illustrates a simplified block-diagram representation of measuring a time interval T_x using the embedded timer of the digital processor (e.g., MCU, FPGA). This measurement is synchronized with the reference clock T_{ref} at the start of the measurement process, which helps reduce the timing error introduced at the beginning (T_{Start}). The digital processor uses an input pin to receive a time interval called T_x . This pin has an input buffer that checks the voltage against a higher threshold (V_{TH}) or a lower threshold (V_{TL}). It must also detect external events using interrupts or a capture module. Each time the input voltage crosses these thresholds, it generates an interrupt, causing the CPU to open or close a digital gate. While the gate is open, the timer counts once for every oscillator period (T_{ref}). This count, called N_x , is stored in a register and represents the digital equivalent of the measured time, expressed as $N_x \approx T_x/T_{ref}$.

In DICs, an embedded timer uses a high-frequency reference oscillator, ideally from outside the digital processor, with period T_{ref} . It converts the RC circuit's discharge time, T_x , into a digital count, N_x . The conversion from time to digital count happens while the capacitor discharges, starting when it has charged to approximately $V_c \approx V_{cc}$. The timer begins counting increments of T_{ref} as long as the capacitor voltage stays above the lower threshold, meaning $V_c > V_{TL}$. When V_c gets close to V_{TL} , an interrupt on the input pin triggers. This stops the timer and saves the count N_x in the capture register, which represents the discharge time as $N_d \approx T_x/T_{ref}$. Figure 10b shows how T_x is converted into N_d .

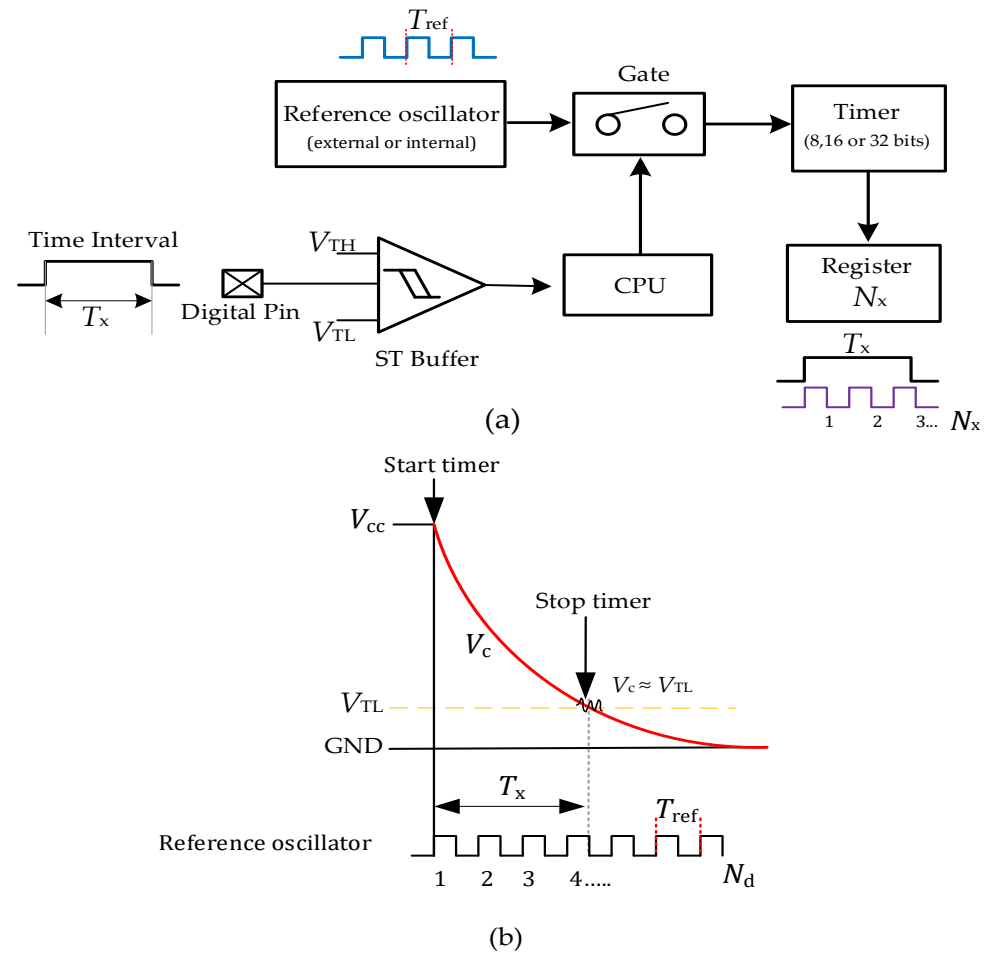


Figure 10. A simplified graphical illustration of (a) a digital processor used for measuring a time interval with its embedded timer, and (b) the conversion of an RC-circuit discharge time into a digital value via the digital processor’s embedded timer.

The time-to-digital conversion faces three main sources of uncertainty [6,37]: (i) instability in the reference oscillator; (ii) quantization errors; (iii) noise during the timer’s trigger or stopping process. Below is a brief description of the effects of these uncertainty sources and the suggested strategies to mitigate them.

- i. **Reference-oscillator instability.** An unstable oscillator introduces variability in the conversion, rendering the measurement of T_x unreliable. The leading causes of oscillator instability include frequency drift (both temporal and thermal) and sensitivity to fluctuations in the supply voltage. To minimize these effects, a crystal oscillator with a total stability of at least ± 50 ppm is advisable, particularly when measuring slowly varying signals (such as an RC discharge) at room temperature over short intervals.
- ii. **Quantization error.** Like an analog-to-digital converter (ADC), time-to-digital conversion is subject to quantization errors. If the timer starts synchronized in firmware, no quantization error occurs at that moment; however, the stop event (trigger) is quantized. The magnitude of this error depends on the detection method (see Table 1). Employing an external interrupt results in a worst-case quantization error for $e_q = M \times T_{ref}$, where M is the number of instruction cycles required to execute the longest interrupt-service routine; employing the capture module limits the quantization error to $e_q = T_{ref}$.
- iii. **Trigger-point noise.** Noise can distort the measured signal or the threshold V_{TL} (see Figure 10b). The primary sources of noise include thermal noise, power supply

noise, and digital processor CPU activity. Recommended countermeasures include placing a decoupling capacitor (as specified by the manufacturer) near the digital processor and using a regulated low-noise power supply. Additionally, putting the digital processor in sleep mode (CPU disabled, timer enabled, interrupts enabled) during the RC-discharge measurement has led to a quantization error $e_q = T_{\text{ref}}$ for both external interrupt and capture module detection methods.

Table 1. Hardware and software techniques used in digital processors for external event detection (timer stop process) during time interval measurement [37].

Technique	Effect
Polling	The CPU periodically polls the logic state of a digital input pin every P instruction cycle. The event is detected in the software. Upon detecting a change on the input pin, the CPU issues a command to start or stop the timer (i.e., to open or close the digital gate). There is no synchronization between the input signal and the reference oscillator. The quantization error is $\pm P \times T_{\text{ref}}$.
General-purpose external interrupt service routines	The CPU either interrupts the execution of the current program or wakes up the MCU when a change (edge or level) occurs on the input pin. Immediately thereafter, it executes the interrupt service routine (ISR), which reads or resets the timer. The quantization error is $\pm M \times T_{\text{ref}}$, where M represents the number of clock cycles required to execute the longest instruction. A dedicated peripheral designed to measure time intervals between two events. A specific pin on the digital processor, linked to this module, operates as an external interrupt. When a transition (edge) is detected on the input pin, the timer's current value is latched (stored) into a dedicated capture register, independently of the executed instruction. Upon completing the current instruction, the CPU branches to the interrupt service routine (ISR), where the captured value is read from the register, not from the timer itself. The quantization error is $\pm T_{\text{ref}}$ (i.e., one-timer count).
Capture module	

2.3. Digital Processor

A digital processor is a programmable electronic device widely used in measurement systems for data acquisition, signal processing, and the execution of complex control algorithms. Its architecture determines its capabilities, with standard implementations based on microcontroller units (MCUs), which typically integrate 8-bit, 16-bit, or 32-bit processing units responsible for managing data and instruction cycles. MCUs include non-volatile Flash memory for storing program code, random-access memory (RAM) for data processing, and various integrated peripherals. These peripherals often include analog comparators, ADCs, digital timers, digital-to-analog converters (DACs), communication interfaces (e.g., USART, SPI, I²C), and general-purpose digital input/output (I/O) pins [38]. An alternative to MCUs is Field-Programmable Gate Arrays (FPGAs). FPGAs consist of configurable logic blocks that can be flexibly interconnected to implement custom digital functions. Like MCUs, they can also incorporate peripheral components, providing a versatile and cost-effective platform for the implementation of complex digital systems [39].

Digital processors play a critical role in DIC implementations by enabling the direct connection of sensors to the processor's digital pins and by leveraging embedded digital timers and counters [6]. These processors typically include a counter or timer module to measure time intervals. An input signal can be applied to a digital pin and compared against voltage thresholds through an internal digital buffer [23]. The processor's Central Processing Unit (CPU) detects voltage crossings via polling or interrupt-driven mechanisms and triggers the counter or timer accordingly. This functionality is commonly based on TTL/CMOS Schmitt trigger inputs, input buffers, or external components, which define a low voltage V_{TL} and a high voltage V_{TH} . The rising edge of the input signal, when it exceeds V_{TH} , marks the start of the time interval, while the falling edge, when it drops

below V_{TL} , defines the end. The counter increments or decrements its value every T_{LSB} seconds, where the digital processor's internal oscillator determines T_{ref} . In this way, the digital processor is responsible for exciting the sensor and processing the DIC signal response using the specific interface technique described in the subsequent sections.

3. Direct Interface Approaches

This section summarizes the fundamental DICs reported in the literature, categorized by the type of circuit used to measure sensors with resistive, capacitive, or inductive characteristics.

3.1. Circuits with RC Components

Reverter et al. [6,7] introduced a basic RC circuit to explain the operational principle of DICs applied to resistive sensors connected directly to an MCU. As shown in Figure 11, the circuit comprises a resistive sensing element R_x and a capacitor C , forming a simple network capable of estimating the sensor's electrical parameters through time-to-digital conversion of the capacitor's charging and discharging processes. In this configuration, the MCU interacts with the circuit using digital input/output pins, denoted as P_{d1} and P_{dP} .

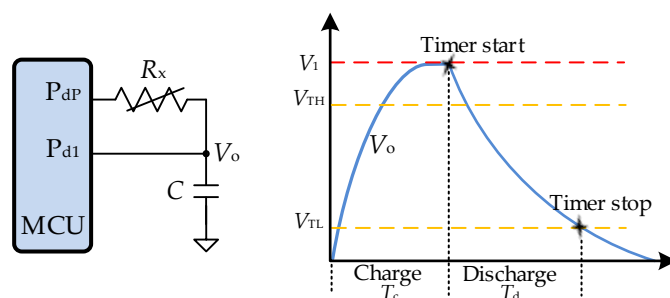


Figure 11. Basic DIC between resistive sensor and MCU showing charge/discharge timing.

The principle of operation relies on measuring the time interval required to charge (T_c) or discharge (T_d) the capacitor C through the resistive element R_x , until a voltage threshold is reached—either the upper threshold V_{TH} or lower threshold V_{TL} , depending on the operation mode. This time interval serves as the basis for estimating the resistance or capacitance of the sensor [7]. This type of DIC can be conceptually interpreted as a classical RC circuit. Assuming the capacitor C is initially discharged (i.e., at 0 V), and a voltage stimulus V_1 is applied to the circuit input, the transient voltage response across the capacitor can be analyzed using the exponential function described as

$$V_o(t) = V_1 \left(1 - e^{-t/RC}\right) \quad (1)$$

and the time required to charge C from 0 V to a high threshold voltage V_{TH} as

$$T_c = RC \ln \left[\left(\frac{V_1}{V_1 - V_{TH}} \right) \right] \quad (2)$$

proportional to R , and if C is charged to a voltage level V_1 , it can be directly discharged to ground from its entry point as is shown in Figure 11, resulting in a transient response in its output voltage $V_o(t)$ as

$$V_o(t) = V_1 \cdot e^{-t/RC} \quad (3)$$

Therefore, the period to discharge C from V_1 to a low threshold voltage V_{TL} can be estimated by using (4):

$$T_d = RC \cdot \ln(V_1/V_{TL}) \quad (4)$$

This implies that any variation in the sensor's resistance is directly proportional to the measured time interval [26]. Consequently, the measurement process is executed in the digital processor in two distinct stages: charging and discharging of the capacitor C . Table 2 summarizes this process, detailing the states of the digital pins and the intervals required to complete each stage.

Table 2. MCU pin configuration and measurement procedure of a resistive sensor.

Stage	P _{d1}	P _{dP}	Process
1	'1'	'HZ'	$T_c = R_i^* C \cdot \ln[V_1 / (V_1 - V_{TL})] \approx 5R_i C$
2	'HZ', and capture time	'0'	$T_d = R_x C \cdot \ln(V_1 / V_{TL})$

* R_i improves noise immunity and is not explicitly shown in Figure 11.

During this operation, the embedded digital timer measures the time taken for the capacitor voltage to cross the lower threshold voltage V_{TL} of the input buffer connected to the input pin P_{d1} (as described in Section 2.2). The resulting digital count is directly proportional to the resistance value R_x . This methodology allows for the conversion of analog resistance into a time-domain signal, which can be digitized and processed by the MCU to estimate R_x , as described by Equation (4).

A second resistor, typically labeled R_i , is placed between P_{d1} and C (Figure 11) to improve rejection of power-supply noise and interference [7]. However, this comes at the expense of increased charging time. The low-pass cutoff frequency defined by R_i and C must be minimized while keeping a charging interval below 1 ms (e.g., if $C = 2.2 \mu\text{F}$, then $R_i < 100 \Omega$).

Similarly, Reverter et al. [6,7] introduced an RC circuit to explain the operational principle of DICs used in capacitive sensors, as shown in Figure 12. The circuit consists of a capacitive sensing element C_x and a reference resistor R .

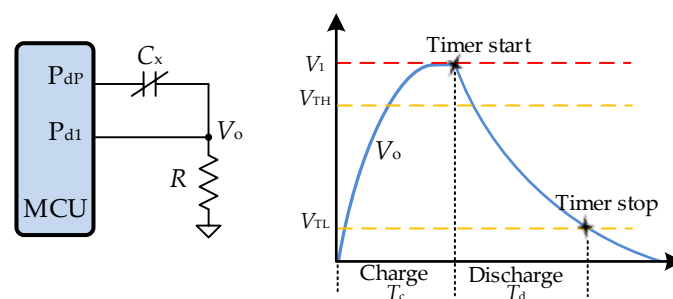


Figure 12. Basic DIC between capacitive sensor and MCU showing charge/discharge timing.

Similarly to the measurement process explained for resistive sensors, this DIC consists of two distinct stages performed in the digital processor. Table 3 summarizes this process; the charge and discharge timing diagrams are the same as those for resistive sensors, indicating that Equation (4) is valid to estimate C_x .

Table 3. MCU pin configuration and measurement procedure of a capacitive sensor.

Stage	P _{d1}	P _{dP}	Process
1	'1'	'0'	$T_c = R_i^* C_x \cdot \ln[V_1 / (V_1 - V_{TL})] \approx 5R_i C$
2	'HZ', and capture time	'0'	$T_d = R C_x \cdot \ln(V_1 / V_{TL})$

* R_i improves noise immunity and is not explicitly shown in Figure 12.

López-Peña et al. [40] presented a similar approach, following the circuit shown in Figure 12, which conducts a calibration-less DIC for capacitive sensors. The charge and discharge processes are controlled via software, utilizing voltage thresholds V_{TH} , V_{TL} ,

V_1 , and 0 V to enhance accuracy. Following this approach, Hidalgo et al. [41] proposed a modification of the DIC, requiring only two additional resistors without any calibration capacitors, thereby increasing resolution and minimizing relative errors.

While the basic configurations in Figures 11 and 12 permit effective resistive and capacitive measurements, they do not account for inherent offset or zero errors. To address this limitation, various authors have proposed a modification of the basic DIC, incorporating a single-point calibration technique [10,11,13,42,43]. As shown in Figure 13, this enhanced configuration introduces a known calibration resistor R_{c1} , which is measured similarly to the sensor R_x . During the measurement cycle, the charging and discharging stages produce two digital counts: N_x , corresponding to the sensor resistance R_x , and N_{c1} , obtained from the calibration reference R_{c1} . This correction mechanism allows compensation for systematic errors and improves measurement accuracy.

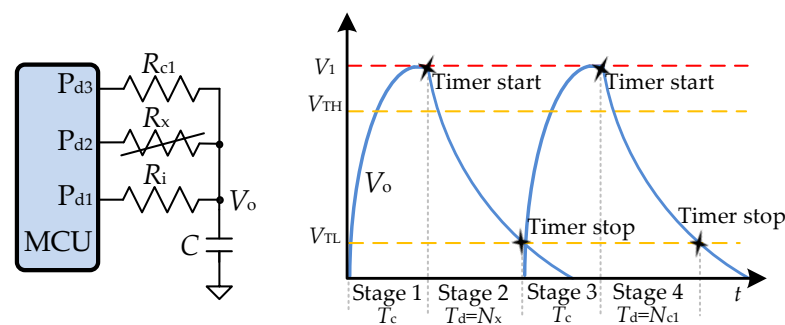


Figure 13. DIC with resistive sensor using single-point calibration and timing process.

This calibration procedure assumes a linear relationship between R_x and N_x , with no zero-offset or non-linearity error requiring further compensation. As a result, both measured values, (R_{c1}, N_{c1}) from the calibration stage and (R_x, N_x) from the sensor, can be used to define a calibration line that passes through the origin (0,0) and the reference point (R_{c1}, N_{c1}) . This linearity allows for the compensation of systematic errors in the measurement process for any R_x value within the range of the calibration point. Accordingly, an estimated value of the sensor resistance R_x can be calculated using Equation (5),

$$R_x = (N_x / N_{c1}) R_{c1} \quad (5)$$

Table 4 summarizes the configuration of the digital processor's pins and the associated measurement procedure used to estimate the sensor resistance R_x through the digital counts N_x and N_{c1} .

Table 4. MCU pin configuration and measurement procedure for single-point calibration of a resistive sensor.

Stage	P _{d1}	P _{d2}	P _{d3}	Process
1	'1'	'HZ'	'HZ'	$T_c \approx 5R_i C$
2	'HZ', and capture N_x	'0'	'HZ'	$N_x = R_x C \ln(V_1 / V_{TL})$
3	'1'	'HZ'	'HZ'	$T_c \approx 5R_i C$
4	'HZ', and capture N_{c1}	'HZ'	'0'	$N_{c1} = R_{c1} C \ln(V_1 / V_{TL})$

Similarly, Richey [11], from Microchip, proposed a DIC for capacitive sensors, incorporating a single-point calibration using a reference capacitor C_{ref} . The circuit is designed to perform two separate measurements: (i) acquisition of the digital count N_x corresponding to the sensor capacitance C_x ; and (ii) acquisition of the digital count N_{ref} from the known reference capacitor C_{ref} , as shown in Figure 14.

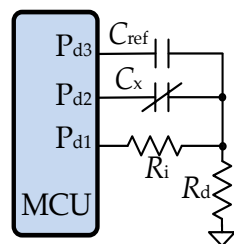


Figure 14. DIC with capacitive sensor using single-point calibration.

The measurement methodology used to estimate C_x in Figure 14 follows the same principle as the resistance estimation technique discussed earlier in Figure 13. However, the pin configuration and measurement setup differ based on the component being measured. Reverter [7,44] also proposed inserting a resistor R_i between P_{d1} and R_d to improve the rejection of power supply noise/interference. The cutoff frequency of the low-pass filter determined by R_i and C_x (during the charging stage) should be kept as low as possible; however, R_i needs to be much smaller than R_d to ensure an adequate charge of C_x (e.g., if $R_d = 1\text{ M}\Omega$, then $R_i < 1\text{ k}\Omega$). Table 5 displays the microcontroller configuration for each measurement stage [45].

Table 5. MCU pin configuration and measurement sequence for single-point calibration of a capacitive sensor.

Stage	P_{d1}	P_{d2}	P_{d3}	Process
1	'1'	'0'	'HZ'	$T_c = 5R_iC_x$
2	'HZ', and capture N_x	'0'	'HZ'	$N_x = R_dC_x \ln(V_1/V_{TL})$
3	'1'	'HZ'	'0'	$T_c = 5R_iC_{ref}$
4	'HZ', and capture N_{ref}	'HZ'	'0'	$N_{ref} = R_dC_{ref} \ln(V_1/V_{TL})$

The sensor capacitance C_x is then estimated using Equation (6), which establishes a direct proportionality between the digital counts N_x and N_{ref} , enabling accurate calibration and compensation.

$$C_x = (N_x/N_{ref})C_{ref} \quad (6)$$

Bierl [12] proposed from Texas Instruments introduces a two-point calibration technique. This method involves the sensor R_x and two calibration resistors, R_{c1} and R_{c2} , as shown in Figure 15. The method is particularly effective when the sensor exhibits zero-offset errors, gain errors, or non-linear tendencies, which cannot be adequately corrected using a single-point calibration.

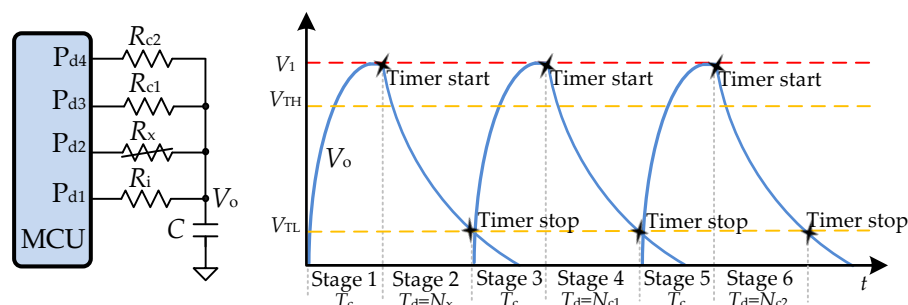


Figure 15. DIC with resistive sensor using two-point calibration and timing process.

To compensate for these effects, the circuit performs three measurement cycles: (i) a digital count N_x obtained from the sensor R_x ; (ii) a reference count N_{c1} measured from the calibration resistor R_{c1} ; and (iii) a second reference count N_{c2} obtained from the calibration

resistor R_{c2} . Each measurement consists of six stages: three for the charging and three for the discharging phases, followed by time-to-digital conversion. Table 6 presents the digital processor configuration for acquiring the three discharge times.

Table 6. MCU pin configuration and measurement sequence for two-point calibration of a resistive sensor.

Stage	P _{d1}	P _{d2}	P _{d3}	P _{d4}	Process
1	'1'	'HZ'	'HZ'	'HZ'	$T_c \approx 5R_iC$
2	'HZ', and capture N_x	'0'	'HZ'	'HZ'	$N_x = R_xC \ln(V_1/V_{TL})$
3	'1'	'HZ'	'HZ'	'HZ'	$T_c \approx 5R_iC$
4	'HZ', and capture N_{c1}	'HZ'	'0'	'HZ'	$N_{c1} = R_{c1}C \ln(V_1/V_{TL})$
5	'1'	'HZ'	'HZ'	'HZ'	$T_c \approx 5R_iC$
6	'HZ', and capture N_{c2}	'HZ'	'HZ'	'0'	$N_{c2} = R_{c2}C \ln(V_1/V_{TL})$

This calibration process enables the construction of a correction curve that passes through the reference points (R_{c1}, N_{c1}) and (R_{c2}, N_{c2}) , effectively compensating for offset, gain, and non-linearity errors. Consequently, the resistance value R_x can be accurately estimated for a given digital output N_x using Equation (7).

$$R_x = [(N_x - N_{c1}) / (N_{c2} - N_{c1})][R_{c2} - R_{c1}] + R_{c1} \quad (7)$$

Hidalgo et al. [46] proposed a method to reduce quantization error based on a two-point calibration technique for resistive sensors, without sacrificing estimation speed or increasing clock frequency. The proposed DIC is shown in Figure 16. The estimation of R_x involves three measurement cycles: (i) a measurement process for R_{c1} ; (ii) a measurement process for $R_{c1} + R_{c2}$; (iii) a simultaneous measurement process through R_x and $R_{c1} + R_{c2}$. Note that R_i performs the same function as described for the single-point calibration technique for capacitive sensors.

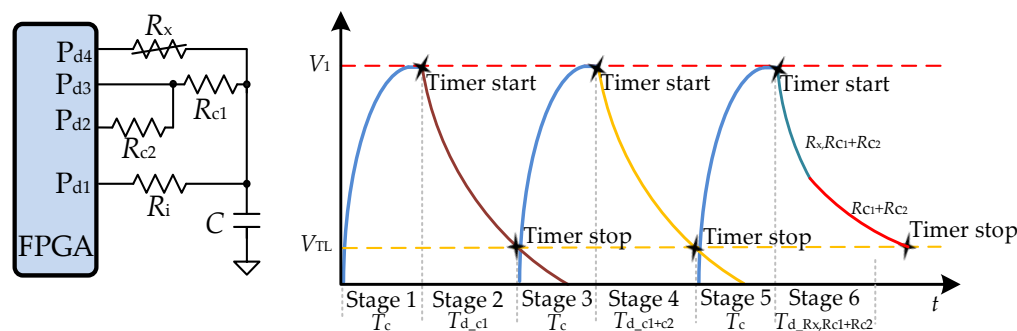


Figure 16. DIC with Quantization Error-Reduction Method for resistive sensors and timing process.

The final value of the sensor resistance R_x can be computed using Equation (8),

$$R_x = [(N_x - N_{c1}) / (N_{c1+c2} - N_{c1})][R_{c2}] + R_{c1} \quad (8)$$

Additionally, Hidalgo et al. [47] also proposed a new configuration for resistive sensors based on two-point calibration, Two-Capacitor Interface (TCI), to decrease trigger uncertainty and reduce errors over the entire measurement range, as shown in Figure 17 (Section 4 explains this proposal in more detail).

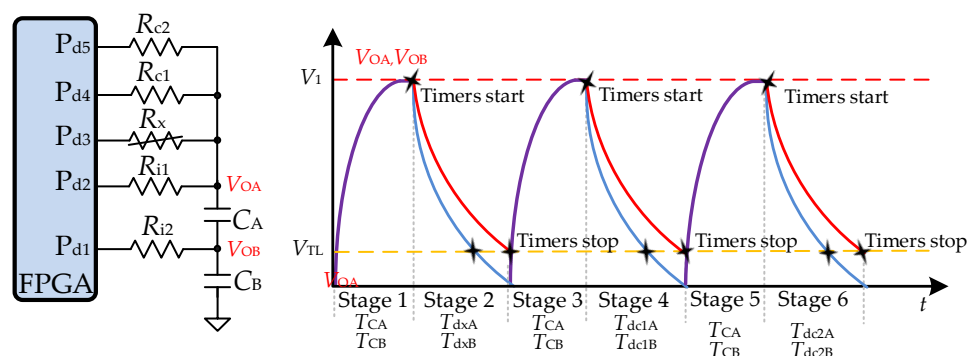


Figure 17. DIC with Two-Capacitor Interface for resistive sensors and timing process.

Van Der Goes et al. [48,49] proposed a two-point calibration technique for capacitive sensors, applied directly to the configuration shown in Figure 14 without additional components. The method utilizes a single known reference capacitor C_{ref} and an open-circuit condition as the second calibration point. This approach facilitates two-point calibration with a minimal hardware setup.

To execute this calibration, the MCU performs three measurements: (i) a digital count from the sensor capacitance C_x ; (ii) a count from the reference capacitor C_{ref} ; and (iii) a counter under open-circuit conditions, which serves as the second reference point C_{off} . Table 7 summarizes the MCU pin configurations for each stage of the measurement process. Once the three measurements are acquired, the sensor capacitance C_x is estimated by Equation (9),

$$C_x = [(N_x - N_{off}) / (N_{ref} - N_{off})][C_{ref}] \quad (9)$$

Table 7. MCU pin configuration and measurement sequence for two-point calibration of a capacitive sensor.

Stage	P _{d1}	P _{d2}	P _{d3}	Process
1	'1'	'0'	'HZ'	$T_c = 5R_i C_x$
2	'HZ', and capture N_x	'0'	'HZ'	$N_x = R_d C_x \ln(V_1 / V_{TL})$
3	'1'	'HZ'	'0'	$T_c = 5R_i C_{ref}$
4	'HZ', and capture N_{ref}	'HZ'	'0'	$N_{ref} = R_d C_{ref} \ln(V_1 / V_{TL})$
5	'1'	'HZ'	'0'	$T_c = 5R_i C_{off}$
6	'HZ', and capture N_{off}	'HZ'	'HZ'	$N_{off} = R_d C_{off} \ln(V_1 / V_{TL})$

Reverter et al. [15] presented a DIC with the three-point calibration technique for resistive sensors, a primary calibration technique proposed by Meijer et al. [50], and taken by Van Der Goes et al. [48,49] for capacitive sensors, which improves upon the previously discussed two-point calibration method. In this approach, the RC circuit shown in Figure 15 is modified by replacing the calibration resistor R_{c1} with a short circuit, as shown in Figure 18.

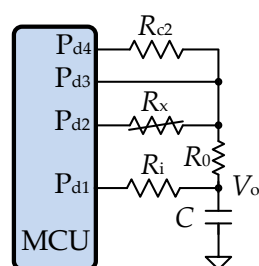


Figure 18. DIC with resistive sensor using three-point calibration technique.

This modification enhances the accuracy in estimating the sensor resistance R_x . In addition, the circuit includes an extra resistor R_0 , which limits the discharge current to the maximum value permitted by the digital processor's output pin. This ensures that the resulting discharge signal retains a well-defined exponential behavior. Since the calibration process is not dependent on the exact value of R_0 , and its temperature-induced variation is negligible, this method provides a more cost-effective alternative for resistance measurement using DICs.

The measurement procedure used to estimate R_x follows the same principle as in the two-point calibration method. Table 8 presents the MCU configuration for each stage of the measurement process. The final value of the sensor resistance R_x can be computed using Equation (10),

$$R_x = [(N_x - N_{c1}) / (N_{c2} - N_{c1})][R_{c2}] \quad (10)$$

Table 8. MCU pin configuration and measurement sequence for three-point calibration of a resistive sensor.

Stage	P _{d1}	P _{d2}	P _{d3}	P _{d4}	Process
1	'1'	'HZ'	'HZ'	'HZ'	$T_c \approx 5R_i C$
2	'HZ', and capture N_x	'0'	'HZ'	'HZ'	$N_x = (R_x + R_0)C \ln(V_1/V_{TL})$
3	'1'	'HZ'	'HZ'	'HZ'	
4	'HZ', and capture N_{c1}	'HZ'	'0'	'HZ'	$N_{c1} = R_0 C \ln(V_1/V_{TL})$
5	'1'	'HZ'	'HZ'	'HZ'	
6	'HZ', Capture N_{c2}	'HZ'	'HZ'	'0'	$N_{c2} = (R_{c2} + R_0)C \ln(V_1/V_{TL})$

3.2. Circuits with RL Components

Kokolanski et al. [21] introduced a DIC for inductive sensors, as shown in Figure 19. This approach utilizes a microcontroller (MCU) with a reference inductor L_{ref} to perform a single-point calibration, following a similar principle to that described in the previous section. An external resistor R_{ext} is incorporated to regulate the current supplied from the digital processor to the sensor inductor L_x .

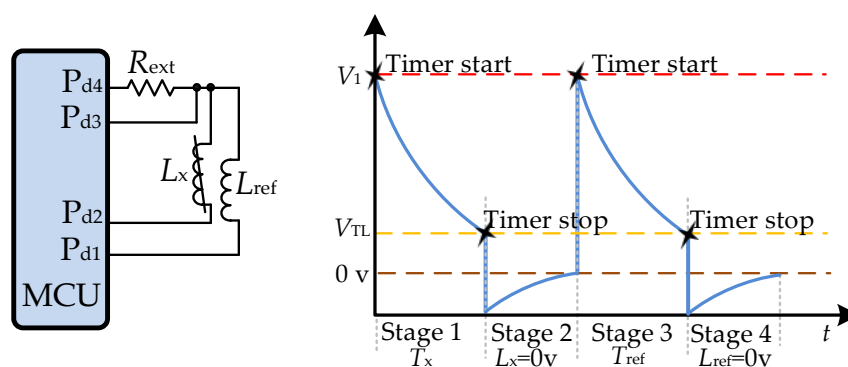


Figure 19. DIC with inductive sensor using single-point calibration and timing process.

The interface consists of two RL circuits configured in a high-pass filter (HPF) topology: one comprising R_{ext} and L_x , and the other consisting of R_{ext} and L_{ref} . Each circuit is energized by the digital processor to measure the discharge times, T_x and T_{ref} , respectively, across the inductors. These time intervals enable the estimation of the sensor inductance L_x . HPF topology is chosen for its ability to provide an extended time constant, which results in longer discharge time and allows the use of smaller values for R_{ext} . Additionally, this configuration leverages the parasitic output resistance of the MCU's digital pin (denoted as P_{d1} in Figure 19), which contributes to the effective series resistance in the RL circuit.

The measurement process involves two separate acquisitions: (i) a measurement from the sensor L_x , yielding the time interval T_x , and (ii) a reference measurement using L_{ref} , producing the interval T_{ref} . Each acquisition is performed through a sequence of four stages. Stages 1 and 2 are associated with the excitation and time measurement of L_x , while stages 3 and 4 are used for the reference inductor L_{ref} . Table 9 outlines the digital processor configurations and measurement sequence. The inductance L_x is then estimated using Equation (11).

$$L_x = (T_x/T_{ref}) L_{ref} \quad (11)$$

Table 9. MCU pin configuration and measurement sequence for single-point calibration of an inductive sensor.

Stage	P _{d1}	P _{d2}	P _{d3}	P _{d4}	Process
1	'1'	'HZ', and capture T_x	'0'	'HZ'	$T_x = L_x/R_{ext} \ln(V_1/V_{TL})$
2	'0'	'HZ'	'0'	'HZ'	Discharge L_x during $5R_{ext} L_x$
3	'1'	'HZ', Capture T_{ref}	'HZ'	'0'	$T_{ref} = L_{ref}/R_{ext} \ln(V_1/V_{TL})$
4	'0'	'HZ'	'HZ'	'0'	Discharge L_{ref} during $5R_{ext} L_{ref}$

3.3. Circuits with Capacitive Charge Transfer

Gaitán-Pitre et al. [51] introduced the operational principle of a DIC based on the capacitive charge transfer technique, which can be analyzed analogously to a traditional RC circuit, as depicted in Figure 20. In this configuration, the sensor capacitance is represented by C_x , the charge transfer capacitor by C_r , and the supply voltage by V_s . These parameters are assumed to remain constant throughout the measurement process.

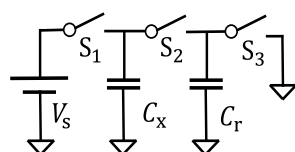


Figure 20. Capacitance charge transfer technique.

The interface operates through the control of a digital processor that counts the number of charge transfer cycles required to raise the voltage across C_r to a predefined threshold level. This method was initially proposed for capacitive sensors in [52,53] and has since been refined to enable accurate digital estimation of C_x . Table 10 outlines the measurement procedure for estimating C_x , consisting of three distinct stages. In each cycle, the charge stored in C_x is transferred to C_r , resulting in a voltage increment across C_r directly proportional to the charge transferred. Repeating the transfer charge increases the voltage across C_r until it reaches V_s , if $C_r > C_x$.

Table 10. The operational principle of capacitive charge transfer.

Stage	S ₁	S ₂	S ₃	Process
1	open	open	close	discharge C_r
2	close	open	open	$T_c = 5RC_x$
3	open	close	open	charge transfer

The total number of transfer cycles N required for C_r to reach the voltage threshold V_{TH} is then measured. Assuming the initial condition $V_r[0] = 0$ V, and considering that $C_r > C_x$, the number of transfer cycles can be estimated using Equation (12). Substituting into the corresponding expression, the sensor capacitance C_x is obtained using Equation (13),

$$N = -C_x^{-1} C_r \ln[(1 - (V_{TH}/V_s))] \quad (12)$$

$$C_x = -\{C_r \ln[1 - (V_{TH}/V_s)]\} / N \quad (13)$$

Dietz [52] presented the first experimental implementation of a DIC based on the capacitive charge transfer technique. The circuit, shown in Figure 21, includes the target sensor capacitance C_x and a reference capacitor C_r .

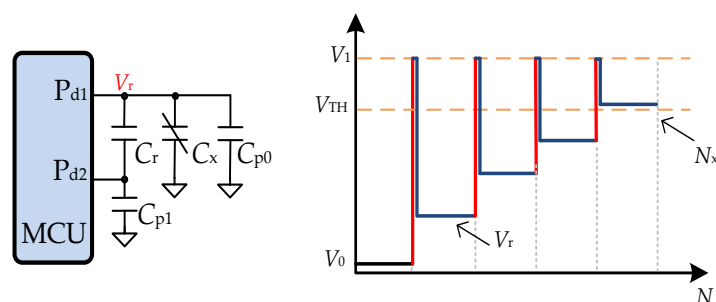


Figure 21. DIC with capacitive sensor using charge transfer technique and timing process.

To account for non-idealities, the design also incorporates parasitic capacitances C_{p0} and C_{p1} , which model the coupling between nodes 0 and 1 for ground and the MCU pins. The circuit further considers the internal output resistance of the MCU's digital buffer, modeled as R_{OL} and R_{OH} , which represent the resistance of the output transistor when a logic level '0' or '1' is applied, respectively. These resistances are assumed to remain constant during operation, provided the transistors function within their ohmic region, as discussed in [15]. The corresponding output voltage levels, V_{OL} and V_{OH} , represent the voltage levels at the MCU pins, respectively, when driven low or high.

The measurement procedure, detailed in Table 11, comprises three sequential stages: circuit initialization, the charging phase, and the charge transfer phase.

Table 11. MCU pin configuration and measurement sequence for charge transfer-based estimation of a capacitive sensor.

Stage	P _{d1}	P _{d2}	Process
1	'0'	'0'	Discharge C_r to $V_r[0] \approx 0$
2	'1'	'HZ'	$T_c = 5RC_x$
3	'HZ', Capture V_{TH}	'0'	Capture of charge transfer cycles N_x

The charge transfer cycles are repeated until the voltage across C_r reaches the predefined threshold voltage V_{TH} of the MCU input buffer. Under the condition that $C_r > C_x$, and considering the parasitic effects from C_{p0} , C_{p1} , and voltage levels V_{OL} and V_{OH} , the number of charge transfer cycles N is recorded. Finally, the capacitance C_x is estimated using (14),

$$C_x = -\{C_r \ln[1 - (V_{TH}/V_s)]\} / N - [C_{p0} + C_{p1}] \quad (14)$$

Gaitán-Pitre et al. [51] introduced an enhanced DIC for capacitive sensing based on the charge transfer method, incorporating a two-point calibration technique designed to make the measurement results independent of V_{OH} , V_{TH} , and C_r . The proposed circuit is shown in Figure 22, and includes only the dominant parasitic capacitances, which are considered the most significant contributors to measurement uncertainty.

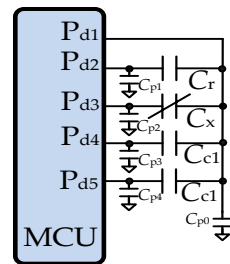


Figure 22. DIC with charge transfer using the two-point calibration technique.

This methodology involves performing three separate measurements: one corresponding to each of the reference capacitors C_{c1} and C_{c2} , and a third for the sensor capacitance C_x . The charge transfer process is applied individually to C_{c1} and C_{c2} to estimate the number of charge transfer cycles required in each case. These measurements yield the digital counts N_x , N_{c1} , and N_{c2} , respectively.

The measurement sequence for each stage follows the same operational procedure previously described and is summarized in Table 11. It is important to note that unused MCU pins must remain configured as high-impedance inputs during each measurement to prevent unintended charge paths. Once all measurements are completed, the sensor capacitance C_x is estimated using Equation (15).

$$C_x = \left[\left(\frac{1}{N_x} \right) \left(\frac{N_{c1}N_{c2}}{N_{c1} - N_{c2}} \right) \right] (C_{c2} - C_{c1}) - \left(\frac{(N_{c2}C_{c2}) - (N_{c1}C_{c1})}{N_{c1} - N_{c2}} \right) \quad (15)$$

4. Results

This section presents a synthesis of the 90 primary studies identified in the review depicted in Section 1.3, classified by circuit category (*RC*, *RL*, and charge transfer capacitive interfaces), sensor type (resistive, capacitive, and inductive), and digital processors (MCU or FPGA). For each study, a uniform set of key performance indicators—including measurement range, accuracy, resolution, acquisition time, systematic errors, and power consumption—enables cross-comparison that reveals prevalent trends, strengths, and limitations as reported in the literature. First, it provides an overview of the key contributions from semiconductor manufacturers through application notes, highlighting the proposed practical implementation strategies. Second, it analyzes the various types of digital processors used in DIC implementations. The rest of the section is structured in thematic subsections: *RC*-based, *RL*-based, charge transfer capacitive interfaces, and applications.

4.1. The Initial Scope of a DIC

The initial development of DICs was driven by semiconductor manufacturers, who introduced the concept through application notes. These early works proposed measuring the electrical value of resistors or capacitors by directly interfacing them with a MCU and measuring the charging or discharging time in an *RC* circuit. The measured time was compared to a fixed threshold voltage, enabling an estimation of the component's value without the need for an analog-to-digital converter (ADC).

Sherman [8], from Philips Semiconductors, demonstrated that using an 80C51 MCU, in conjunction with a timer, a monostable multivibrator, and a comparator, it was possible to establish a strong linear relationship between the charging time of an *RC* circuit and its capacitance or resistance. This work laid the foundation for the fundamental operating principle behind DICs.

Webjör [9] from Motorola Semiconductors proposed a method for measuring resistance in temperature and pressure sensors using the MC68HC05 MCU. The method consisted of

two measurement phases, calibration and sensor acquisition, allowing for improved accuracy through reference point definition, variability compensation, output normalization, and systematic error reduction.

Cox [10], from Microchip, described the implementation of a digital ohmmeter using a PIC16C5X MCU. The technique involved charging and discharging a capacitor to determine a time value associated with the sensor, followed by a calibration process designed to eliminate first-order errors such as offset, gain deviations, capacitance tolerances, power supply fluctuations, and temperature effects. A similar approach was proposed by Richey [11], also from Microchip, using a PIC16C622 MCU and an RC circuit to estimate resistance or capacitance by measuring the capacitor's charging time to a fixed reference voltage, monitored via an analog comparator.

Additionally, Merritt [13] from Texas Instruments introduced a digital thermometer based on an RC circuit and the timers of the MSP430x325 MCU. This implementation obtained sensor values by charging and discharging a capacitor through a thermistor and a reference resistor. This enabled the MCU to count clock cycles and determine the corresponding resistance, which was then converted into temperature.

These application notes illustrate the feasibility of reducing external components in measurement systems by utilizing internal MCU resources. The proposed DIC strategies encourage the idea of direct sensor-to-MCU interfaces as an affordable and compact alternative to traditional analog signal conditioning. These initial developments highlight the potential of MCUs in measurement applications and establish a foundation for a research field that continues to garner considerable academic and industrial interest.

4.2. Digital Processors Used in DIC

In DICs, the digital processor serves as the central computational unit, responsible for interpreting and processing sensor signals. In sensor interfacing applications, it integrates essential functionalities such as data acquisition, control logic, and communication, typically within a single embedded platform. DIC implementations leverage the digital processor's capability to convert sensor responses into measurable time intervals by utilizing internal digital pins and embedded timers.

Table 12 summarizes the digital processors reported in the literature for DIC implementations, categorizing them based on their operating frequency, embedded timer resolution, timer measuring range, and operating voltage range.

Table 12. Summary of digital processors used in DIC implementations, including operating frequency, timer resolution, measuring range, and voltage.

Ref.		Digital Processor	Manufacturer	Clock Reference	Timer-Counter Bits	Time Resolution	Supply Voltage (V)
[34,41,54–56]	FPGA	Artix 7 XC7A35T	Xilinx	50 MHz	18 bits	20 ns	3.3
[46,47,57,58]	FPGA	Spartan 6 (XC6SLX25-3FTG256)	Xilinx	50 MHz	14 bits	20 ns	3.3
[59–61]	FPGA	Spartan 3 (XC3S50AN-4TQG144C)	Xilinx	50 MHz	14 bits	20 ns	1.2–3.3
[62–64]	MCU	ATXmega32A4	Atmel	16 MHz	16 bits	62.5 ns	3.3
[21,35,65,66]	MCU	ATmega328P	Atmel	16 MHz	16 bits	62.5 ns	3.3–5
[16,18–20,31,33,67,68]	MCU	AVR ATtiny2313	Atmel	20 MHz	16 bits	50 ns	5
[15,37,69,70]	MCU	AVR AT90S2313	Atmel	4 MHz	16 bits	250 ns	5
[22]	MCU	SAM3X8E ARM Cortex-M3	Atmel	84 MHz	32 bits	Prescaler (11.9 ns × 2) = 23.8 ns	3.3
[15,44,45,71,72]	MCU	PIC16F873	Microchip	20 MHz	16 bits	200 ns	5
[59]	MCU	PIC16LF1559	Microchip	32 MHz	16 bits	125 ns	3.3
[73]	MCU	PIC16F876	Microchip	20 MHz	16 bits	200 ns	5
[21]	MCU	PIC16F877A	Microchip	5 MHz	16 bits	800 ns	5

Table 12. Cont.

Ref.	Digital Processor	Manufacturer	Clock Reference	Timer-Counter Bits	Time Resolution	Supply Voltage (V)	
[74,75]	MCU	PIC16F877A	Microchip	20 MHz	16 bits	200 ns	5
[76]	MCU	PIC18F45K22	Microchip	20 MHz	16 bits	200 ns	5
[52]	MCU	PIC12C508	Microchip	not specified	8 bits	Not specified	5
[51,53,77,78]	MCU	PIC16F84A	Microchip	4 MHz	16 bits	1 μs	5
[79]	MCU	PIC18F4680	Microchip	8 MHz	16 bits	125 ns	5
[80]	MCU	MSP430F1471	Texas Instruments	32 KHz	16 bits	7 MHz (DCO) = 142 ns	3.3
[40]	MCU	MSP430F1471	Texas Instruments	8 MHz	16 bits	125 ns	3.3
[81–83]	MCU	MSP430F123	Texas Instruments	8 MHz	16 bits	125 ns	3.3
[17]	MCU	MSP430F123	Texas Instruments	4 MHz	16 bits	250 ns	3.3v
[84]	MCU	MSP430F2274	Texas Instruments	8 MHz	16 bits	125 ns	3.3
[85]	MCU	STM32F401CBU6	ST Micro	48 MHz	16 bits	20.83 ns	2.7–3.6
[86]	MCU	STM32L073RZT6	ST Micro	64 MHz	16 bits	15.6 ns	1.65–3.6
[87]	MCU	D1 mini ESP32	Espressif Systems	80 MHz	64 bits	12.5 ns	3.3
[88]	MCU	C8051F040	Silicon Labs	25 MHz	16 bits	40 ns	2.7–5.6
[89]	FPGA	Cyclone III EP3C10	Intel	48 MHz	16 bits	20 ns	3.3

As reflected in Table 12, MCUs constitute the most used digital processors in DIC systems. However, in recent years, FPGAs have gained relevance due to their higher processing capabilities and timing precision.

MCU implementations typically operate within a frequency range of 4 MHz to 32 MHz. Notable exceptions include implementations at 84 MHz [22], 80 MHz [87], 64 MHz [86], and 42 MHz [85], which demonstrate that increased clock frequency directly enhances the resolution of the embedded timer. Most MCUs have 16-bit timer-counters, while some high-performance devices, such as the SAM3X8E [22] and the D1 Mini ESP32 [87], offer 32-bit and even 64-bit timers, respectively. This expanded bit-width accurately measures longer time intervals and improves overall timing resolution.

FPGAs, by contrast, operate at significantly higher frequencies and offer enhanced timing resolution, often as precise as 20 ns, due to their configurable logic and high-speed clocks. Lower-frequency MCUs typically exhibit time resolutions ranging from 125 ns to 800 ns. Only a few MCU-based implementations achieve finer resolutions, in the range of 12.5 ns to 23.8 ns, and these are directly associated with devices running at higher clock speeds.

This analysis indicates that while MCUs dominate DIC applications due to their integration and cost efficiency, FPGAs provide superior precision and performance for high-resolution time measurement applications. Additionally, when comparing FPGAs and MCUs in a DIC implementation, the decision often involves balancing key trade-offs in flexibility, performance, power consumption, and cost. In contrast, FPGAs offer unparalleled flexibility for hardware design and exceptional performance for parallel tasks, which MCUs cannot match. However, this flexibility and performance come at an increased cost and higher power consumption. Meanwhile, MCUs provide an easier programming experience using high-level languages like C/C++, making them more accessible for general-purpose applications. FPGAs require expertise in hardware description languages (HDLs) like VHDL or Verilog. FPGAs are generally more expensive than MCUs but allow for extensive hardware customization. MCUs, on the other hand, have fixed architectures, limiting their ability to adapt to unique or complex requirements.

It suggests that the choice of an FPGA or MCU depends on the specific needs of the application (e.g., IoT sensor interfaces, Industry 4.0, and wearable smart sensors), such as the level of performance required, power efficiency, development resources, and cost.

4.3. DIC with RC Components

DICs based on RC components are the most widely adopted approach for sensor interfacing, particularly for sensors that exhibit resistive or capacitive variations. Numerous

studies have been dedicated to improving accuracy, resolution, and measurement time and reducing sources of uncertainty and interference that impact system performance. Table 13 highlights the most relevant contributions and associated data collected for further analysis.

Table 13. Summary of DIC implementations with RC components for resistive and capacitive sensors.

Ref.	Sensor Type	DIC Components	Keywords	Operative Parameters Reported
[33]	Four-wire resistive sensors	Three resistors, a capacitor, a switch, a sensor, and an MCU.	A direct approach for interfacing four-wire resistive sensors.	• Measurement ranges from 60 Ω to 220 Ω. • Time measurement approx. 10 ms. • Non-linearity errors around 0.05% FSS.
[31]	Three-wire resistive sensors	Three resistors, a capacitor, a sensor, and an MCU.	A DIC for three-wire connected resistive sensors.	• Measurement ranges from 60 Ω to 264 Ω. • Time measurement approx. 7 ms. • Non-linearity errors around 0.02% FSS.
[34]	Simple resistive sensors	Two resistors, a capacitor, a sensor, and an FPGA.	Two-Measurement Method (TMM) for resistive sensors affected by lead-wire resistances.	• Measurement ranges from 100 Ω to 2000 Ω for lead-wire resistances from 0 Ω to 100 Ω. • Time measurement approx. 1.09 ms. • Non-linearity errors were around 0.12% FSS.
		Three resistors, a capacitor, a sensor, and an FPGA.	Improved Method (IM) for resistive sensors affected by lead-wire resistances.	• Measurement ranges from 100 Ω to 2000 Ω for lead-wire resistances from 0 Ω to 100 Ω. • Time measurement approx. 1.09 ms. • Non-linearity error around 0.15% FSS.
[68]	Non-linear resistive sensors, such as thermistor B57164K from TDK	Three resistors, a capacitor, a sensor, and an MCU.	DIC for non-linear resistive sensors.	• Measurement ranges from 5 kΩ to 30 kΩ. • Time measurement approx. 1.3 ms per cycle. • Power consumption reported at 30 μJ. • Non-linearity error less of 1% FSS.
[54]	Differential resistive sensors	Two capacitors, a resistor, a sensor, and an FPGA.	Proposal of a simple digital readout DIC for differential resistive sensors.	• Measurement ranges from 0.442 kΩ to 23.5 kΩ. • Time measurement from 1.1 ms to 1.3 ms. • Resolution obtained around 11.4 bits. • Measurement errors at 0.34%.
	Differential capacitive sensors	Two resistors, a capacitor, a sensor, and an FPGA.	Proposal of a simple digital readout DIC for differential capacitive sensors.	• Measurement ranges from 0.5 nF to 34.9 nF. • Time measurement around 1.3 ms. • Resolution obtained around 10.8 bits. • Measurement errors were around 0.63%.
[16]	Differential resistive sensors for linear position, angular position, pressure sensors, or level sensors	Two resistors, a capacitor, a sensor, and an MCU.	Interfacing differential resistive sensors with DIC.	• Measurement ranges to emulate a sensor of 1.0 kΩ to (−1, +1). • Time measurement approximately to 1 ms for the largest capacitor value. • Resolution obtained around 7.4 bits to 11.6 bits. • Non-linearity error around 0.01% FSS.
[19]	Differential capacitive sensors, such as the accelerometer SCG10Z-G001CC from VTI Technologies	Two resistors, a sensor, and an MCU.	Interfacing differential capacitive sensors with DIC.	• Measurement ranges of sensors of 0.039 nF (−0.07, +0.07). • Time measurement around 50 ms. • Resolution obtained around 7 bits. • Non-linearity error around 1% FSS.
[55]	Simple resistive sensors include resistance temperature detectors (RTD), gas, force, or humidity sensors	Two capacitors, a sensor, and an FPGA.	Proposal of Two-Capacitor Interface (TCI) to simplify resistive sensor readout.	• Measurement ranges from 0.221 kΩ to 24.9 kΩ. • Time measurement around 1.2 ms. • Relative errors around 0.2 to 0.3%. • Power consumption reported at 476 nJ.
		Two resistors, a sensor, and an FPGA.	Proposal of Single-Capacitor Interface (SCI) to simplify resistive sensor readout.	• Measurement ranges from 0.221 kΩ to 24.9 kΩ. • Time measurement around 0.685 ms. • Relative errors around 0.16 to 0.7%. • Power consumption reported at 476 nJ.

Table 13. Cont.

Ref.	Sensor Type	DIC Components	Keywords	Operative Parameters Reported
[41]	Simple capacitive sensors, such as liquid level, pressure, strain, and humidity sensors	Two resistors, a sensor, and an MCU.	A proposal of a new DIC to simplify the capacitive sensor.	• Measurement ranges from 0.1 nF to 561 nF. • Time measurement around 0.12 ms for each 1 nF reading. • Relative errors were around 0.41 to 1.01%. • Linearity errors ranging from 0.03% to 0.3%.
[56]	Capacitive sensors	Two resistors, an operational amplifier, a sensor, and an FPGA.	DIC for capacitive sensors affected by parasitic series resistances.	• Measurement ranges from 100 pF to 96 nF. • Time measurement around 4.5 ms. • Systematic error of about 0.37% when parasitic series resistances are as high as 1200 Ω.
[15]	Simple resistive sensors, such as Pt1000 RTD sensors	Two resistors, a capacitor, a sensor, and an MCU.	A study of the accuracy and resolution of a DIC for resistive sensors.	• Measurement ranges from 0.800 kΩ to 1.5 kΩ. • Time measurement around 2.372 ms. • Relative errors around 0.01%. • Maximum resolution was obtained around 0.10 Ω to 0.30 Ω. • Power consumption reported at 768 nJ.
[47]	Simple resistive sensors, focused on tactile sensors	Four resistors, two capacitors, a sensor, and an FPGA.	A proposal of a Two-Capacitor Direct Interface Circuit (TCDIC) to enhance the measurement of resistive sensors.	• Measurement ranges from 0.128 kΩ to 7.4 kΩ. • Time measurement around 1.05 ms. • Maximum absolute error in estimating Rx from 13.4 Ω to 14.3 Ω. • Power consumption increased slightly from 4.8 to 6.5% compared with the two-point calibration technique.
[64]	Lossy capacitive sensors	Two resistors, a capacitor, a sensor, a comparator, and an MCU.	A proposal of a DIC designed for lossy capacitive sensors, such as humidity sensors.	• Measurement ranges from 1 kΩ up to 10 MΩ. • Time measurement approximately to 0.7 ms for each measurement cycle. • Relative uncertainty less than 0.74% with systematic errors are maintained within approximately $\pm 0.5\%$. • Power consumption achieved to 8.73 mA at 3.3 V.
[46]	Simple resistive sensors include thermistors, strain gauges, pressure sensors, and biomedical sensors	Three resistors, two capacitors, a sensor, and an FPGA.	A proposal of a Quantization Error-Reduction Method (QERM) to reduce quantization errors when measuring low resistance values.	• Measurement ranges from 0.033 kΩ to 8.169 kΩ. • Maximum relative errors of approximately 1.56% for low resistances.
[57]	Simple resistive sensors	Three resistors, a capacitor, a sensor, and an FPGA.	Proposal to reduce time measurement in DIC for resistive sensors.	• Measurement ranges from 0.270 kΩ to 7.5 kΩ. • Time measurement around 0.002 ms for each measurement cycle. • Errors reported between 0.2% to 0.3% for most resistance values.
[63]	Simple capacitive sensors, as capacitive relative humidity (RH) sensors	A resistor, an analog comparator, a DAC, a sensor, and an MCU.	A proposal of a measurement method based on a versatile DIC with internal MCU peripherals.	• Measurement ranges from 0.1 nF to 0.225 nF. • 64 cycles resulting in a measurement time of 61.2 ms to 126.5 ms. • Accuracy obtained around 0.1 pF. • Maximum relative error of 0.06%.
[59]	Simple resistive sensors include temperature, gas, anemometers, and tactile sensors	Four resistors, a capacitor, a sensor, and an MCU.	MCU proposes an Improved Calibration Method (ICM) to provide more accurate measurements and reduce relative errors.	• Measurement ranges from 0.010 kΩ to 7.5 kΩ. • Maximum relative error at 5.5%.
		Four resistors, a capacitor, a sensor, and an FPGA.	FPGA proposes an Improved Calibration Method (ICM) to provide more accurate measurements and reduce relative errors.	• Measurement ranges from 0.0099 kΩ to 6.7 kΩ. • Maximum relative error at 3.0%.

Table 13. Cont.

Ref.	Sensor Type	DIC Components	Keywords	Operative Parameters Reported
[66]	Capacitively coupled resistive sensors	A reference resistor, a capacitor, an operational amplifier, capacitively coupled resistive sensors, and an MCU.	A first approach of a DIC for capacitively coupled resistive sensors.	- Measurement ranges from 50 kΩ to 800 kΩ, and capacitance specified at ranges around some pF. - Time measurement around 10 ms. - Maximum relative errors at 0.91% for resistance measurements and 2.94% for capacitive measurements.
[60]	Simple resistive sensors include thermistors, gas detection, magneto-resistive, and tactile sensors	Two resistors, a capacitor, a sensor, and an FPGA.	A proposal of two fast calibration methods for faster, more efficient, and more accurate sensor data acquisition.	- Measurement ranges from 0.260 kΩ to 9.96 kΩ. - Time measurement around 0.413 ms to 0.553 ms.
[42]	Resistive sensors	Three resistors, a capacitor, a sensor, and an FPGA.	Quasi single point calibration method for high-speed measurements.	- Measurement ranges from 267 Ω to 7.46 kΩ. - Time measurement reduction of 61% against those in [43].
[90]	Capacitive sensors	Digital switches, an hysteresis voltage comparator circuit, a single voltage comparator, sensor, and an MCU.	An enhanced method for measuring capacitance with analog components to correct threshold voltages.	- Measurement ranges from 100 pF to 2200 pF. - Isolated digital pins with switches.
[62]	Simple resistive sensors as thermistors or strain gauges.	A resistor, a capacitor, an ADC, an analog comparator, two MOSFETs, a sensor, and an MCU.	The time-domain measurement method for resistive sensors is based on a versatile DIC with external components.	- Measurement ranges from 0.100 kΩ to 8.2 kΩ. - Measurement time for each cycle at 7.25 μs, a complete measurement requires 64 cycles to ensure accuracy. - Resolution of 12 bits. - A relative error of less than 3%.
	Simple capacitive sensors, such as touch sensing, pressure sensing, liquid level measurement, or proximity sensors		The time-domain measurement method for capacitive sensors is based on a versatile DIC with external components.	- Measurement ranges from 100 nF to 12,000 nF. - Measurement time for each cycle at 7.25 μs, a complete measurement requires 64 cycles to ensure accuracy. - Resolution of 12 bits. - A relative error of less than 0.2%.
[35]	Simple resistive sensors include RTD, thermistors, LDRs, strain gauges, gas sensors, and piezoresistive sensors	Two resistors, a capacitor, an analog comparator, two diodes, a sensor, and an MCU.	Propose a DIC that minimizes measurement errors due to lead wire resistance variations and temperature compensation.	- Measurement ranges from 145.96 Ω to 146.42 Ω. - Time measurement around 5.3 ms. - Resolution of 12 bits with an approximate sensitivity of 0.03 Ω.
[80]	Simple resistive sensors	Two resistors, a capacitor, a sensor, and an MCU.	A proposal for a low-power consumption DIC through a design guide for the selection of optimal parameters related to the measurement setup for resistive sensors.	- Measurement ranges from 1 kΩ to 1.38 kΩ, and 100 Ω to 138 kΩ. - Time measurement around 1.5 ms. - Effective number of bits from 9 bits to 12 bits. - Power consumption from 1.9 μJ to 5 μJ for the resistance values given.
[40]	Simple capacitive sensors	A resistor, a sensor, and an MCU.	Proposal of calibration-less DIC.	- Measurement ranges from 47 nF to 220 nF. - Accuracy was obtained around 1 pF. - Relative errors from 0.1 to 2% across various capacitor values.

Table 13. Cont.

Ref.	Sensor Type	DIC Components	Keywords	Operative Parameters Reported
[67]	Simple resistive sensors	A resistor, a capacitor, a sensor, and an MCU.	Analysis for optimizing DICs to enhance their power consumption, particularly for battery-operated applications.	• Measurement ranges around 1 kΩ. • Time measurement from 1 ms to 6 ms. • Resolution of 13 bits. • A non-linearity error of 0.01% FSS. • Current consumption in active mode for measuring a 1 kΩ resistive sensor approximately at 1.5 mA.
	Simple capacitive sensors	Two resistors, a sensor, and an MCU.		• Measurement ranges from 10 pF to 1 nF, 1 nF to 100 nF, and few μF. • Time measurement from 1 ms to 6 ms. • Resolution of 9 bits. • A non-linearity error of 0.1% FSS. • Current consumption in active mode for measuring a 177 nF capacitive sensor approximately at 0.6 mA.
[20]	Lossy capacitors, such as P14–Rapid capacitive humidity sensor	Two resistors, a capacitor, a sensor, and an MCU.	Proposal DIC for lossy capacitance sensors.	• Measurement ranges from 0.150 nF to 0.206 nF. • Time measurement is around 2 ms to 3 ms. • Relative errors of 0.3% for Gx = 100 nS and 6.0% for Gx = 1 μS.
[81]	Simple resistive sensors, such as Pt1000 temperature sensors or magneto-resistive sensors	Two resistors, a capacitor, a sensor, and an MCU.	Proposal to improve traditional DIC performance using Vernier technique.	• Measurement ranges around 1 kΩ. • Time measurement around 0.2 ms. • Accuracy obtained around 0.114 Ω.
[17]	Bridge type resistive sensors, such as the full-bridge HMC1052 sensor, an Anisotropic Magneto-resistive (AMR) sensor or the half-bridge AAH002 sensor, a Giant Magneto-resistive (GMR)	A resistor, a capacitor, a sensor, and an MCU.	A DIC proposal to linearize resistive sensor bridges.	• Measurement ranges from 1 kΩ to 2 kΩ. • Measurement time is around 1 ms per cycle, and the full-bridge proposal requires three measurement cycles. • Effective number of bits of 11 bits. • Full-bridge circuit non-linearity errors around 0.2% FSR. • Half-bridge circuit non-linearity errors around 0.3% FSR.
[71]	Bridge type resistive sensors, such as MPXV53GC7U piezoresistive pressure sensors	A resistor, a capacitor, a Schmitt trigger, a sensor, and an MCU.	A simple and efficient DIC proposal for piezoresistive pressure sensors.	• Measurement ranges are not specified; instead, the pressure measurement range is presented as being from 0 to 7.5 psi. • Three measurement cycles are required for Req1, Req2, and Req3. • Maximum non-linearity error at 1.5% FSS, averaging 0.5% FSS.
[18]	Simple capacitive sensors, like Philips H1 Sensor and Humirel HS1101 Sensor	Two resistors, a capacitor, a sensor, and an MCU.	A DIC proposal for capacitive humidity sensors.	• Measurement ranges from 99 pF to 146 pF, and 149 pF to 206 pF. • 100 measurement cycles for the sensor readings with an overall measuring time of approximately 50 ms. • Sensitivity varies from 0.2 pF to 0.5 pF per %RH. • Effective number of bits of 9 bits. • Non-linearity errors of 0.11% FSS.
[45]	Simple capacitive sensors	A resistor, a capacitor, a sensor, and an MCU.	A low-cost DIC proposal for low-value capacitive sensors.	• Measurement ranges from 10 pF to 100 pF. • Absolute error was reported below 1.5% FSR.
[43]	Simple resistive sensors	Two resistors, a capacitor, a sensor, and MCU.	One-point auto-calibration technique sensor interface.	• Measurement ranges around 1 kΩ to 100 kΩ. • Measurement time under μs ranges for resistances around 1 kΩ, and tens of μs for ranges 100 kΩ. • Non-linearity errors of 1.3% FSS.

Table 13. Cont.

Ref.	Sensor Type	DIC Components	Keywords	Operative Parameters Reported
[91]	Resistive sensor arrays	A sensor array, a capacitor, and an FPGA.	DIC for resistive sensor arrays: resolution analysis.	• Measurement ranges around 200 Ω to 7.35 kΩ. • Measurement time with a single resistor and 500 estimation, the standard deviations ranged from approximately 13.58 ns for the 199.96 Ω resistor to about 38.46 ns for the 7348.84 Ω resistor. • Effective resolution achieved approximately to 10.14 bits.
[44]	Analysis of power-supply interference effects on DIC sensor-MCU			
[69]	Effective number of resolution bits in direct sensor-to-microcontroller interfaces			
[37]	Uncertainty reduction techniques in microcontroller-based time measurements			
[61]	Innovative capture modules for DIC sensor-FPGA			
[70]	Measurement error analysis and uncertainty reduction for period and time interval-to-digital converters based on MCU			

Reverter [33] presented a simple DIC for four-wire resistive sensors using an external single switch. This approach reduces the complexity and number of components required compared to traditional methods that often involve multiple switches and external circuitry, achieving a maximum non-linearity error at 0.05% FSS, and relatively rapid measuring time (approximately 10 ms) for low resistance values associated with sensors like the Pt100 temperature sensor, in a resistance range from 60 Ω to 220 Ω .

According to [92], the Reverter contrasts an improvement where the sensor interface needs four external switches, where measurement time is directly affected around 60 ms.

Reverter [31] also proposed a DIC for three-wire resistive sensors. The circuit requires more charge–discharge cycles and a longer conversion time. Additionally, two-wire resistances are expected to be equal; otherwise, there is an offset error. However, it is noted for requiring fewer external components than existing solutions, thereby reducing overall complexity and cost while maintaining measurement accuracy. Reverter presented measurement ranges from 60 Ω to 264 Ω with a non-linearity error of around 0.03% and 0.02% FSS, which enabled improvements over [32,93], where multiple comparators, operational amplifiers, and switches are needed to perform a three-wire sensor measurement.

Hidalgo [34] introduced two novel DICs for the digital readout of resistive sensors to mitigate the errors introduced by lead wire resistances by obtaining multiple time measurements during a single capacitor charge and discharge. The Two-Measurement Method (TMM) circuit utilizes two resistors and a capacitor to acquire two-time measurements to estimate the sensor's resistance. The Improved Method (IM) includes an additional resistor for a third time measurement, which reduces errors in estimating the resistance. Experimental results were obtained by configuring known resistances to ensure their combined resistance is significantly larger than the lead wire resistances, effectively diminishing their impact on sensor readings. Hidalgo presented experimental results with sensor measurements at 100 to 2000 Ω for lead-wire resistances from 0 to 100 Ω , achieving significant measurement accuracy with systematic errors as low as 0.12% for TMM and 0.15% for IM. Both proposals effectively address the challenges of lead-wire resistance in measuring resistive sensors, representing an improvement of [94], where measurements ranges from 1000 to 1100 Ω for lead-wire resistances from 0 to 10 Ω , achieved significant measurement accuracy with systematic errors at 0.33%.

Reverter [68] presented DIC as capable of measuring non-linear resistive sensors, such as thermistors. The proposed circuit employs an autocalibration and linearization process by incorporating a reference resistor RCL in parallel with R_x , and the non-linear resistance of the thermistor is transformed into a more linear response. Reverter presented experimental results of the proposal circuit with a thermistor B57164K from TDK, with

resistive variations approximately 5 k Ω to 30 k Ω . After hardware linearization, non-linearity reported to be less than 1% FSS. This approach conducts a highly integrated, low-cost, energy-efficient, and accurate measurement technique, with simplicity, auto-calibration, and adaptability benefits.

Areekath et al. [66] presented a DIC for capacitively coupled resistive sensors, which has not been explored so far from the perspective of a DIC. These sensors are suitable for non-contact sensing applications such as water quality monitoring or other environmental measurements. Areekath et al. presented experimental results with measurement ranges from approximately 50 k Ω up to 800 k Ω , corresponding to typical water conductivity monitoring, such as measuring water quality. On the other hand, the capacitance measurement range is not explicitly specified. However, it is evaluated in the order of a few picofarads, with experimental results showing maximum errors under 3% across these ranges. According to measurement time, the proposed DIC completes both resistance and capacitance measurements in approximately 10 ms. Additionally, the maximum error reported is about 0.91% for resistance measurements and 2.94% for capacitive measurements. This paper contributes to exploring alternatives for capacitively coupled resistive sensors with DICs.

Hidalgo [54] presented a circuit designed for differential resistive and capacitive sensors, achieving measurement times of between 1.1 ms and 1.3 ms and non-linearity errors of 0.34% Full Scale Span (FSS) for resistive sensors and 0.63% FSS for capacitive sensors. These results correspond to practical resolutions of 11.4 and 10.8 bits, respectively. Compared to previous works by Reverter et al. [16,19], this study demonstrated significant performance improvements. However, the method's disadvantage lies in the relatively long acquisition time per cycle, which may limit fast-monitoring applications.

Hidalgo et al. [47,55] introduced two simplified DICs for resistive sensors by reconfiguring the two-point calibration method described in [15], enabling resistance estimation through a single charge–discharge cycle. This approach resulted in a more compact configuration, shorter measurement time, and up to a 40% reduction in energy consumption. Hidalgo et al. proposed two new configurations for resistive sensors based on RC networks: the Two-Capacitor Interface (TCI) and the Single-Capacitor Interface (SCI). These approaches utilize only two digital pins, reduce power consumption, and achieve faster acquisition times compared to traditional two-point calibration techniques. The TCI configuration decreases measurement time by approximately 50%, while the SCI configuration attains a 75% reduction for resistance ranges between 221 Ω and 24.9 k Ω . Although no resolution values were provided, the study emphasizes the relationship between resolution and equivalent capacitance, indicating that higher capacitance enhances resolution, albeit with potential increases in uncertainty due to the discharge curve characteristics.

Czaja [64] proposed a DIC method for lossy capacitive sensors, such as humidity sensors. The author emulated the lossy capacitive sensor with capacitance values ranging from 100 pF to 285.85 pF and resistances from 1 M Ω to 10 M Ω , ensuring accurate testing and calibration of the measurement system. This proposal is optimized to expedite measurements compared to traditional approaches, facilitating quicker sensor readings with lower energy consumption, and a systematic error correction algorithm applied, using a correction dictionary with multiple measurements to improve accuracy with relative uncertainty less than 0.74% and systematic errors approximately to $\pm 0.5\%$ within a time measurement approximately to 0.7 ms for each measurement cycle.

Hidalgo et al. [41] developed an innovative DIC for capacitive sensors that eliminates the need for calibration capacitors by using only two additional resistors and a Xilinx Artix 7 FPGA (XC7A35T). This configuration estimates the capacitance with a single charge and discharge cycle. Although resolution data is not reported, the authors indicate a maximum

relative error of 1.01% and an average error of 0.41% across a range from 0.100 nF to 561 nF, with an estimated acquisition time of 0.12 ms per 1 nF.

Hidalgo [56] presented a DIC for capacitive sensors affected by parasitic series resistances to obtain reliable capacitance measurements in a single charging-discharging cycle, with advantages in minimizing power consumption and eliminating errors caused by parasitic resistances, making it highly suitable for IoT and ambient intelligence applications. Hidalgo presented experimental results measuring capacitances from approximately 100 pF to 96 nF, maintaining as low as possible systematic error of about 0.37% within this range, even when parasitic series resistances are as high as 1200 Ω .

Hidalgo et al. [46] introduced the Quantization Error-Reduction Method (QERM) to address quantization errors, which modifies the discharging process by simultaneously discharging the sensor and calibration resistors. This approach significantly reduces noise quantization and improves overall resolution. The method requires two calibration resistors, a load resistor, a capacitor, and an FPGA. For resistance measurements between 33 Ω and 8.169 k Ω , the maximum error for values near 33 Ω was 0.33 Ω , representing a 1.56% relative error, in contrast to the 3.77% error observed using the standard two-point calibration technique.

Hidalgo et al. [57] proposed the Short-Time Charging Calibration Method (SCCM), which shortens the charging phase and allows two measurements per discharge cycle, significantly reducing acquisition time. Compared to the two-point calibration method, SCCM reduced charge times from 40.96 μ s to 1.92 μ s, with a modest increase in error (0.2% to 0.3%) attributed to quantization effects at shorter time intervals, suggesting the alternative of increasing the clock frequency of the FPGA or the capacitance values. However, this may lead to higher power consumption or increased noise.

Further contributions by Czaja [63] involved using an ATXmega32A4 MCU with an embedded DAC and comparator to measure capacitance in humidity sensors, achieving errors below 0.07 pF and relative errors under 0.06% for experimental measurement ranges between 100 pF and 225 pF. This method involved averaging 64 measurements to improve accuracy, making it highly suitable for low-power sensor networks.

Hidalgo et al. [59] presented the Improved Direct Interface Circuit (IDIC), which uses three calibration resistors to build a linear system for estimating resistance. Implemented on a PIC16LF1559 MCU and a Xilinx Spartan3AN FPGA, the IDIC outperformed the two-point method, reducing the error from 26.6% to 5.5% for resistances near 80.388 Ω . The maximum reported error with FPGA implementation was 3%. In [60], two fast calibration methods (FCM I and II) were proposed, which use the smallest calibration resistor during discharge to reduce measurement time. These methods achieved conversion times between 0.413 ms and 0.553 ms and were up to 55% faster than the two-point calibration technique. The reported measurement ranges span from 260 Ω to 9.963 k Ω , with measurements between 0.413 ms and 0.553 ms, conversion times up to 55% faster, with consistent errors for different resistance values up to 2.198 k Ω . However, FCM II introduced higher errors at larger resistance values.

Botín-Córdoba et al. [42] presented a DIC named quasi single-point calibration methods (QSPCMs) for resistive sensors that significantly reduce measurement time, maintaining acceptable accuracy. This proposal uses an initial calibration with two known resistors to establish baseline parameters, and subsequent measurements by measuring only one resistor, and a calibration process with the other. These methods incorporate accelerated discharge procedures as fast calibration methods [60]. These QSPCMs achieve a reduction of approximately 61% against those in [60] for the mean estimation time when performing 20 measurements across the resistance range of 267.56 Ω to 7.46 k Ω .

Czaja [62] proposed a mixed time-domain DIC using an ATXmega32A4 MCU, passive components, and a MOSFET-based switching network to measure resistance and capacitance. The method uses a voltage divider with a reference resistor to estimate sensor resistance, averaging 64 measurements and applying piecewise linear corrections. Resistance errors remained under 3% from 100 Ω to 8.2 k Ω . Capacitive measurement errors were below 0.2% for values between 100 nF and 12 μ F. Nagarajan et al. [35] addressed cable-induced resistance errors using an ATmega328 MCU, reducing measurement error to 0.06% for a Pt100 sensor. Their design allowed remote resistive sensors to be connected directly to the MCU without significant precision loss.

Wang et al. [90] proposed an enhanced method for measuring capacitance, which employs a constant voltage source to charge the capacitive sensor, voltage comparators to stabilize the fluctuating threshold voltages, and incorporates analog switches to isolate the different discharging circuits completely. The circuit measures capacitance in the range from 100 pF to 2200 pF, using the timing of charge–discharge cycles by voltage comparators. Errors from voltage fluctuations and parasitic capacitances are minimized through circuit enhancements, achieving improved accuracy, the method provides precise capacitance measurements with limited uncertainties.

López et al. [80] developed a low-power DIC achieving 9-bit ENOB at 5 μ J for 1–1.38 k Ω and 1.9 μ J for 100–138 k Ω . López et al. [40] also presented a capacitor calibration-free DIC using only a reference resistor and the MSP430F1471 MCU. The design maintained uncertainties below 1 pF for capacitances between 33 pF and 4.7 nF.

Reverter [67] analyzed current consumption for resistive and capacitive DICs, reporting 1.5 mA and 0.6 mA, respectively, due to shorter charging times in capacitive sensors. Reverter et al. [20] addressed lossy capacitive sensors by correcting parasitic conductance using a calibration technique that maintains accuracy even in high-humidity environments. The design achieved a relative error of less than 1%.

Sifuentes et al. [81] introduced a Vernier-based method to extend discharge time and enhance resolution without increasing power consumption. Jordana et al. [71] and Sifuentes et al. [17] implemented DICs for bridge-configured resistive sensors [95], achieving non-linearity errors of 0.4% to 1.5% FSR and validating measurement accuracy through time averaging.

Oballe-Peinado et al. [91] presented a development and experimental validation of a DIC using a FPGA for resistive sensor arrays. This approach eliminates resistive crosstalk and simplifies hardware by balancing sensing speed and precision, offering a practical and scalable solution for tactile sensing and other applications that involve resistive sensor arrays. The authors reported experimental results showing maximum relative errors as low as 0.066% for resistances ranging from approximately 200 Ω to 7.3 k Ω .

Oballe-Peinado et al. [96] presented an enhanced DIC employing operational amplifiers with capacitive feedback to mitigate errors introduced by non-idealities of operational amplifiers and FPGA input/output drivers and applying calibration algorithms that effectively reduce crosstalk effects. These advancements enable larger measurement ranges with higher precision.

Kokolanski et al. [43] introduced a sensor interface and experimental validation of a continual one-point auto-calibration technique for DICs, improving the measurement range and reducing non-linearity errors without needing multiple calibration resistors. Experimental results are given for resistive sensors in the range of 1 k Ω to 100 k Ω , with the measured discharge times on the order of microseconds, generally spanning from a few microseconds for low resistances around 1 k Ω up to tens of microseconds for higher resistances around 100 k Ω .

Reverter et al. [37,44,45,69] examined the effects of power supply interference, ENOB, and quantization uncertainty. The use of input capture modules significantly improved resolution, while general-purpose interrupts led to increased timing uncertainty. Oballe-Peinado et al. [61] proposed an FPGA-based smart capture module that filters noise and enhances timing precision. Yurish et al. [70] emphasized the importance of oscillator stability, noting that clock sources can introduce uncertainty ranging from ± 50 ppm to ± 100 ppm depending on the oscillator type.

4.4. DIC with RL Components

The implementation of DICs with *RL* components is primarily oriented toward measuring sensors that exhibit variations in inductance. This approach targets sensor–microcontroller interfaces for inductive sensors. Table 14 highlights this principle’s most relevant publications and experimental results.

Table 14. Summary of DIC implementations with *RL* components for inductive sensors.

Ref.	Sensor Type	DIC Components	Keywords	Operative Parameters Reported
[21]	Simple inductive sensors for displacement or position measurement, and inductive pressure or temperature sensors.	A resistor, an inductor, a sensor, and an MCU.	A DIC proposal for low-pass filter (LPF) inductive sensors.	• Measurement ranges from 1.0 mH to 10.0 mH. • Resolution obtained around 10.5 bits. • Non-linearity errors are lower than 0.3% FSS.
[74]	Simple inductive sensors.	A resistor, an inductor, four MOSFETs, a sensor, and an MCU.	Improving the resolution in a DIC for LPF and high-pass filter (HPF) inductive sensors.	• Measurement ranges from 1.0 mH to 9.0 mH. • Time measurement around 0.1832 ms. • Effective number of bits improved by 0.6 bits compared to the basic HPF configuration and nearly 2.2 bits over the LPF topology.
[22]	Differential inductive sensors include linear variable differential transformer (LVDT) and differential variable reluctance transducer (DVRT).	A resistor, an analog comparator, an inductor, two diodes, a sensor, and an MCU.	A simple and effective digitizer designed for differential variable inductive and reluctance sensors.	• Measurement ranges from 6.0 mH to 50.0 mH. • Time measurement around 0.200 ms. • Relative errors in a commercial LVDT range of 0–25 mm at 0.77% during measurements across a displacement range of 15 mm in 1 mm increments. • Relative errors in a commercial DVRT also demonstrated a worst-case error of less than 0.83% when measuring displacement in steps of 1 mm.
[62]	Simple inductive sensors.	A resistor, a capacitor, an ADC, an analog comparator, two MOSFET, a sensor, and an MCU.	Time-domain measurement method for inductive sensors based on a versatile DIC with external components	• Measurement ranges from 0.1 mH to 30.0 mH. • Time measurement should be around 1 ms per cycle, and 64 cycles are needed to ensure accuracy. • Resolution achieved around 12 bits. • Relative errors of inductance at 0.3%, and in some cases, even less than 0.1% for the inductive sensor.
[97]	Simple inductive sensors.	A resistor, an inductor, an analog comparator, a sensor, and an MCU.	DIC proposal for resolution enhancement for inductive sensors.	• Measurement ranges from 0.1 mH to 1.0 mH. • Resolution achieved in terms of accuracy at 4 μH. • Maximum non-linearity errors are less of 0.4% FSS, while the non-linearity error for the lower range stays around 1.1% FSS due to quantization effects. • Maximum current sourcing/sinking capacity of DIC below 40 mA.
[98]	Differential inductive sensors, like the SM260.100.2 inductive sensor from Schreibe, for linear displacement.	A resistor, a sensor, and an MCU.	DIC proposal for differential inductive sensors.	• Measurement ranges in inductance are not provided, instead proposal can measure linear displacements up to ± 50 mm. • Non-linearity error was around 1% FSS. • Resolution achieved around 9 bits.
[99]	Investigation of errors in the microcontroller interface circuit for the mutual inductance sensor.			

Kokolanski et al. [21] introduced the first DIC for inductive sensors using a reference RL circuit based on RC circuit principles. The proposed system employed a PIC16F877A and an ATmega328P MCU, integrating a reference inductor and a current-limiting resistor to form two RL circuits. Using a single-point calibration method, the system achieved inductance estimation in the range of 0.01 mH to 1.0 mH, with an effective resolution of 10.5 bits and a non-linearity error (NLE) below 0.3% FSS.

Kokolanski et al. [74] improved the original design by incorporating external MOSFETs to reduce the overall resistance in the circuit, thereby increasing the time constant of the RL network. This enhancement led to more accurate measurements by extending the discharging period. As a result, an additional 0.6 bits of resolution were achieved compared to the high-pass filter (HPF) topology, and up to 2.2 bits compared to the low-pass filter (LPF) configuration. The system also controlled power consumption through an external resistor, limiting the peak current to 100 mA, with a 47 Ω resistor selected for the HPF setup.

Ramadoss et al. [22] proposed a DIC based on the ATSAM3x8E MCU for applications involving differential inductive sensors such as Linear Variable Differential Transformers (LVDTs) and Differential Variable Reluctance Transducers (DVRTs). The system achieved displacement measurement ranges of 0–25 mm for LVDTs and 0–10 mm for DVRTs. Two measurements per inductor segment were performed, with total measurement times up to 200 ms. The reported error of 0.77% was primarily attributed to temperature variations and resistor mismatches, demonstrating suitability for industrial applications.

As previously mentioned in Section 4.3, Czaja [62] developed a time-domain measurement technique applicable to resistance, inductance, and capacitance sensors. For inductive measurements, the experimental setup yielded relative errors of less than 0.3% across a range of 0.1 mH to 30.0 mH, highlighting its potential as a low-cost and energy-efficient method for smart sensor systems.

Asif et al. [97] proposed a method to improve resolution in inductive DICs without external analog-to-digital converters (ADCs). The technique involved stepwise supply voltage modulation using either an MCU pin (ATmega328P) or an external source, in combination with resistance optimization. The circuit measured voltages during charge and discharge phases to identify inductive response and estimate inductance values. The setup achieved a resolution of 4 μH over a 0.01–1 mH range, with a maximum non-linearity error between 1.1% and <0.4%, corresponding to an ENOB of 7.9. The system maintained low power consumption with a peak current of less than 40 mA.

Kokolanski et al. [98] demonstrated a practical displacement sensing application using DIC and a single inductive element with an ATmega328P MCU. The system used a Vernier Caliper to validate displacements in the ± 50 mm range. The circuit achieved a resolution of approximately 9 bits and non-linearity errors of 1%. Both HPF and LPF topologies were evaluated, demonstrating a reduction in complexity, cost, and power consumption compared to conventional analog interfaces.

Anarghya et al. [99] addressed error sources in inductive DIC implementations, identifying parasitic resistance and capacitance from MCU I/O pins and inductors as contributors to signal distortion and measurement instability. The presence of trigger noise and quantization effects during time-to-digital conversion also degrades accuracy. The authors suggested future improvements in shielding, grounding, and digital noise mitigation techniques, particularly for linear displacement measurement using mutual inductance sensors.

4.5. DIC with Capacitive Charge Transfer

The charge transfer technique is implemented between a sensor and MCU to estimate the sensor value by counting the number of charge transfer cycles needed to raise the voltage across a reference capacitor. This methodology enables compact sensor interfacing

using only a digital processor and a few passive components. Table 15 highlights relevant publications and experimental data associated with this technique.

Table 15. Summary of DIC implementations using capacitive charge transfer technique.

Ref.	Sensor Type	DIC Components	Keywords	Operative Parameters Reported
[52]	Simple capacitive sensors, fluid level monitoring.	A capacitor, a diode, an inductor, a sensor, and MCU.	Wireless liquid level sensing for restaurant applications with charge transfer technique.	Measurement ranges in this work indicate level sensing with a resolution of 0.1%.
[51,53]	Simple capacitive sensors.	Two capacitors, a sensor, and MCU.	DIC proposal for simple capacitive sensors with charge transfer technique.	- Measurement ranges from 0.01 nF to 0.1 nF, and 0.1 nF to 1.0 nF. - Measurement time is around 1000 ms. - A maximum measurement deviation around 0.01% FSR for capacitance measurements in the range of 10 to 100 pF and from 100 pF to 1 nF and 0.08% FSR for the subrange from 2 pF to 10 pF. - External calibration ranges from 10 pF to 100 pF with a maximal deviation of around 0.015% FSR when using external calibration.
[77]	Differential capacitive sensors measure various physical quantities, such as linear or angular position, displacement, pressure, and force.	Two capacitors, a sensor, and MCU.	DIC proposal for differential capacitive sensors with charge transfer technique.	- Measurement ranges from 0.1 nF to 1.0 nF. - A maximal relative deviation was achieved at $\pm 4\%$ FSS for ranges of 100 pF and a significantly lower relative deviation of $\pm 0.6\%$ FSS for ranges of 1 nF. - Measurement time for each transfer charge cycle is around 10.030 ms.
[78]	Simple resistive sensors, such as NTC Thermistors or LDR.	Three resistors, two capacitors, a sensor, and MCU.	DIC proposal for simple resistive sensors with charge transfer technique.	- Measurement ranges from 0.100 kΩ to 10.0 MΩ. - Measurement time for each transfer charge cycle is around 10.275 ms at a range of 100 kΩ to 1 MΩ. - Measurement time for each transfer charge cycle is around 12.525 ms at 1 MΩ to 10 MΩ. - A maximal relative errors of $\pm 4\%$ FSS in the range of 100 kΩ to 1 MΩ, and maximal relative errors of $\pm 5\%$ FSS in the range of 1 MΩ to 10 MΩ.
[72]	Analysis of capacitive interferences in DIC with the transfer charge technique.			

Dietz et al. [52] presented the first practical DIC employing the charge transfer technique, which was used for detecting liquid levels in restaurant glassware. The proposed circuit estimates the fluid level by counting the number of charge transfer cycles required to charge a reference capacitor to a given digital threshold voltage. However, a resolution of up to 0.1% was reported.

Gaitán-Pitre et al. [51] introduced the operational principle of a two-point calibration technique for capacitive charge transfer using a PIC16F84A MCU, with a complementary analysis found in [53]. The method improves accuracy by compensating for parasitic capacitances, voltage offsets, and temperature-induced effects. It involves measuring two known reference capacitors and the target capacitance sensor, thereby correcting systematic errors and reducing gain uncertainty. Although power consumption was not quantified, the authors emphasized that using a low-cost MCU and design techniques for minimizing interference can result in energy-efficient implementations. While exact measurement times were not specified, each charge transfer cycle was estimated to take approximately 1030 ms, potentially limiting its applicability in time-sensitive scenarios.

Gaitán-Pitre et al. [77] introduced a DIC tailored for differential capacitive sensors using the same MCU platform. The design supports capacitance measurements ranging from 0.1 nF to 1 nF, with each charge transfer cycle lasting approximately 10.030 ms. The system reported maximum relative deviations of $\pm 4\%$ for nominal capacitances of 100 pF and $\pm 0.6\%$ for 1 nF. Suggested applications include linear and angular position sensing,

displacement, pressure, and force measurements, making them suitable for research and industrial environments. Gaitán-Pitre et al. [78] extended the application of the charge transfer technique to high-resistance sensors, ranging from 100 k Ω to 10 M Ω . This design employed a reference capacitor, a transfer capacitor, and a PIC16F84A MCU. The system achieved accuracies of $\pm 4\%$ for resistance values between 100 k Ω and 1 M Ω , and $\pm 5\%$ between 1 M Ω and 10 M Ω . Its strong immunity to external capacitive interference and adaptability to sensor arrays make it ideal for high-impedance sensor networks requiring cost-effective, low-complexity solutions.

Gaitán-Pitre et al. [72] conducted an in-depth analysis of capacitive interference in DICs employing the charge transfer method. The results demonstrated that the technique maintains low equivalent impedance (approximately 3 k Ω), making it less vulnerable to interference than RC-based circuits. Additionally, the study showed that calibration via charge transfer can effectively mitigate parasitic capacitance effects, while RC techniques tend to experience increased measurement deviations as parasitic capacitance rises. Therefore, the charge transfer technique is well suited for measuring very small capacitances, particularly in electromagnetically noisy environments.

DICs based on capacitive charge transfer provide a robust and compact solution for measuring capacitive and resistive sensors, particularly when high resolution and a low component count are desired. Although some limitations exist concerning measurement speed, especially for high-precision applications, these systems are well-suited for low-power, noise-resilient sensing in embedded environments.

4.6. Summary of DIC Applications

Table 16 summarizes various sensor interfaces and their reported applications, illustrating their relevance in health monitoring, environmental sensing, and safety systems. For instance, researchers have implemented a DIC to investigate how aging affects temperature readings from resistive temperature sensors. They have also examined the impact of Bias Temperature Instability (BTI) and proposed a novel calibration method using a reference resistor and switching mechanism to reduce BTI-related errors throughout a system's lifespan [100].

Table 16. A summary of DIC applications reported in the reviewed literature.

Ref.	Sensor Type	Application
[100]	Resistive temperature sensor	Impact of aging on temperature measurements
[86]	Resistive temperature sensor	Accuracy of an NTC thermistor measurements
[85]	Light emitting diodes (LEDs) for light sensing	Monitoring vegetation health and density
[87]	Force sensing resistor	Heart rate measurement
[82]	Force sensing resistor	Seat occupancy detection system
[58]	Force sensing resistor	Application into tactile sensing systems, robotic applications, and other force or pressure sensing environments where precise force measurement is critical
[65]	Magneto-resistive sensor	Measuring dynamic signals such as Electrocardiogram (ECG) signals
[76]	Metal oxide semiconductor (MOS) gas sensor	Gas sensors are used to monitor and discriminate between different gases
[101]	Chemoresistive gas sensor	Gas sensor to estimate CO and NO ₂
[83]	Nasal thermistor and piezoresistive sensor	Application in clinical and home settings for respiratory rate detection systems for health monitoring
[88]	Grounded cylindrical capacitive sensor	Water level monitoring for tanks
[89]	LEDs for a photoplethysmography (PPG) sensor	Oximeter for blood oxygen saturation monitoring
[84]	Giant magneto-resistive (GMR) sensor and an LDR	Wireless magnetic sensor node for vehicle detection

Table 16. Cont.

Ref.	Sensor Type	Application
[18]	Capacitive humidity sensor	Humidity sensing for industrial automation
[73]	Grounded capacitive sensor	Liquid-level measurement for industrial processes
[52]	Capacitive sensor	Wireless liquid level sensing
[21]	Inductive sensor	Displacement and position measurement
[22]	Differential inductive sensors	Linear variable differential transformer (LVDT), and differential variable reluctance transducer (DVRT)
[79]	Piezoresistive tactile sensors	Humanoid robots, industrial robotics, medical prosthetics, and rehabilitation devices
[75]	RFID capacitive humidity sensor	RFID labels for wireless sensing, such as food traceability and industrial monitoring
[102–104]	Universal Sensors and Transducers Interfacecircuit (USTI)	An integrated circuit for resistive, resistive-bridge, and capacitive sensors, such as measurement of gases, humidity, temperature, pressure, displacement, and biomedical devices
[105]	Ultrasonic smart sensors	Distance measurement with a USTI integrated circuit for distance measurements, tank level measurements, object detection and monitoring, garage parking assistance, and motion detection systems

Another recent study aims to accurately measure temperature using an NTC 3950 thermistor, demonstrating the potential and benefits of deploying low-cost, energy-efficient temperature sensing solutions in portable and wearable IoT devices [103].

An interesting approach is the vegetation monitoring using LEDs for NDVI estimation [85], wearable technologies employing force sensing resistors (FSRs) for heart rate detection [87], and vehicular safety systems leveraging FSRs for seat occupancy monitoring [82], and tactile sensing systems, robotic applications, and other force or pressure sensing environments where precise force measurement is critical [58].

Other applications range from ECG acquisition using magneto-resistive sensors [65], gas detection with MOS sensor arrays [76], gas detection with a chemoresistive sensor to estimate CO and NO₂ [101], respiratory monitoring through combined thermistor and piezoresistive sensors [83], and water level control with grounded capacitive sensors [88].

Additional implementations include photoplethysmography using LEDs [89], smart city vehicle detection via GMR and LDR sensor integration [84], and humidity sensing with capacitive sensors for automation and climate control [18]. Liquid level detection in industrial and hospitality settings has also been demonstrated using grounded capacitive sensors and capacitive glassware sensing solutions [52,73]. Inductive sensors for displacement and position monitoring [21], and distance measurement with LVDT AND DVRT [22]. An innovative application is the development of advanced tactile sensing systems for robotics and assistive devices that enable more accurate, real-time force sensing in robotic fingers, palms, or skin-like surfaces for humanoid robots, industrial robotics, medical prosthetics, and rehabilitation devices [79]. Pelegrí-Sebastiá et al. [75] introduced a low-cost and low-power capacitive humidity sensor into flexible RFID labels designed for wireless sensing and tracking in various environments, such as food traceability and industrial monitoring, allowing real-time monitoring of relative humidity without adding significant cost or power consumption to the RFID tag, facilitating applications where environmental conditions need to be tracked and recorded remotely and efficiently.

Yurish [102–104] introduced a complete solution, an integrated circuit based on DICs concepts for resistive and capacitive sensors, the so-called Universal Sensors and Transducers Interface (USTI), for measuring gases, humidity, temperature, pressure, displacement, and biomedical devices. Additionally, Yurish [96] presented complete application for a distance measurement system based on ultrasonic smart sensors using USTI integrated circuits.

Hidalgo [106] introduced sigma-delta techniques in a DIC to read resistive sensors, focusing on enhancing accuracy, reducing measurement time, and cutting energy use. Sigma-delta ($\sigma\text{-}\Delta$) is a popular analog-to-digital conversion technique that provides high-resolution measurements with simplicity and noise-shaping capabilities. It operates by oversampling the analog signal at a frequency much higher than the Nyquist rate and employing a feedback loop with an integrator, comparator, and digital filter. The proposed DIC introduces two parameters— M (number of measurement cycles) and N (clock cycles charging capacitor)—which allow the optimization of the trade-off between accuracy, acquisition time, and energy consumption according to application needs in experimental ranges from 211 Ω to 16 k Ω . This proposal ideal for applications in portable devices, biomedical devices, Industry 4.0, and the IoT.

5. Discussion

This paper offers a detailed overview of DICs utilizing RC , RL , and capacitive charge transfer configurations. Figure 4 illustrates a general summary of DIC advancements over the years, as outlined in Table 17. Additionally, DIC contributions can be represented in a pie-to-pie chart by DIC configuration, as shown in Figure 23. This chart offers a detailed breakdown of their primary contributions, allowing for a deeper exploration of each study’s key insights and advancements.

Table 17. DIC advancements over the years.

Decade	Percentage	Contribution
1990s	10%	Basic concepts, application notes, and simple DICs; publications specifically focused on detailed interface designs.
2000s	21%	Operative principles of DIC, significant improvement papers, such as calibration techniques to increase accuracy and reduce uncertainty errors, and analysis papers of key parameters.
2010s	42%	Most advances in compact circuits, error mitigation techniques, improved calibration methods, low-power solutions, and DIC applications.
2020s	27%	Novel sensor interfaces for non-linear sensors, correction of parasitic resistance effects with wire-resistance techniques for resistive, capacitive, and remote sensors.

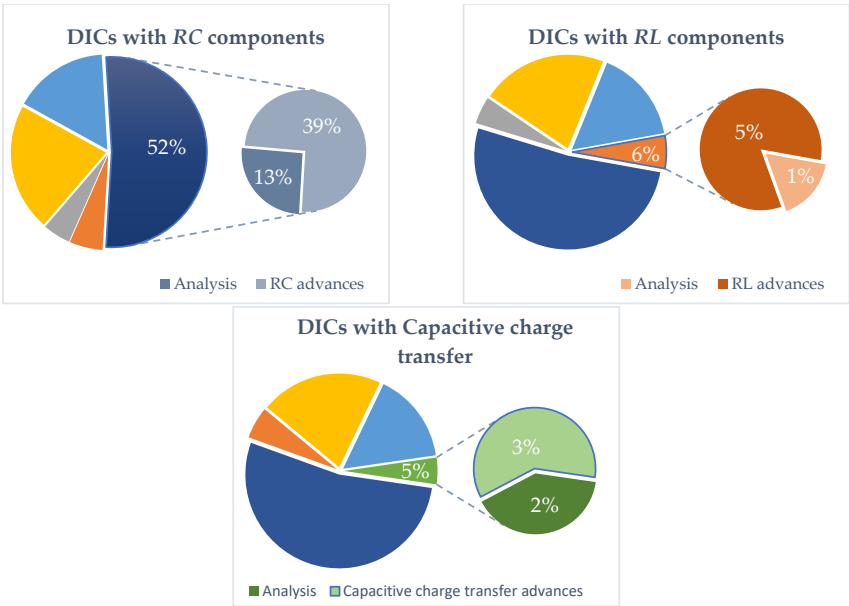


Figure 23. Percentage contribution area for each DIC studied.

5.1. DIC with RC Components Discussion

DICs with RC components are examined in two areas: implementation analysis and recent advances. Implementation analysis accounts for 13% of related topics, covering operating principles [6,7,25,26,46,49,95], accuracy and resolution [15,37,69,91], power consumption [67], power supply effects and interferences [44], and overall performance [81]. Advances, 39% of the literature, include interfacing simple capacitive and resistive sensors [8–13,35,40–43,46–48,55,57–60,62–64,71,80,86,90,92,93], differential resistive sensors [16,54], resistive sensor bridges [17,95], differential capacitive sensors [19,54], lossy capacitive sensors [20,64], low-value capacitive sensors [45], four-wire resistive sensors [33], three-wire resistive sensors [31,32], sensors affected by lead-wire resistance [34,94], non-linear resistive sensors [68], capacitively coupled resistive sensors [66], capacitive sensors influenced by parasitic series resistances [56], and resistive sensor arrays [96].

According to Section 4, accuracy limitations are critical because they affect measurement reliability and precision. These limitations originate from systematic errors, environmental factors, and inherent design constraints. A general overview follows:

- **Systematic errors.** Quantization errors, trigger noise, and non-linearity are common limitations. Quantization particularly affects low-resistance measurements, yielding higher relative errors. Calibration techniques mitigate these errors: employing sleep mode and capture modules reduces noise [37], decoupling capacitors reduce power-supply interference in quasi-digital sensors [44]. Recent innovations, such as the Two-Capacitor DIC [47], reduce systematic errors in high-resistance measurements by up to 90% compared to conventional DICs. Other methods (e.g., Three-Signal Auto-Calibration [15,48,50] and Quasi Single-Point Calibration [42]) improve resistive sensor accuracy but often require additional components or resources. For instance, the Improved Calibration Method (ICM) [59] lowers error for low-resistance measurements at the cost of greater system complexity.
- **Trade-offs.** DIC designs balance accuracy, response time, and energy consumption. High-accuracy DICs typically consume more power due to advanced calibration or high-frequency clocks [67]. Energy-efficient designs reduce power consumption but may compromise accuracy and precision [52]. Fast calibration methods (FCMs) for resistive sensors [60] cut measurement times by up to 55% but introduce slightly higher errors than the Two-Point Calibration Method (TPCM). Likewise, sigma-delta DICs [106] improve accuracy by increasing measurement cycles (M) at the expense of longer acquisition times and higher energy use. Aging compensation techniques enhance long-term accuracy but require additional calibration cycles or hardware [100].
- **Parasitic components and non-idealities.** Parasitic capacitances and resistances introduce significant error [30,34,63], especially in capacitive and resistive interfaces. Stray capacitances can produce up to 0.37% error in capacitive sensor readouts. Resistive sensor arrays face crosstalk due to multiplexing and amplifier non-idealities, limiting accuracy. MCU and FPGA input impedances also matter; interfaces without Schmitt trigger buffers increase uncertainty [61]. Additionally, certain sensor types have inherent limitations: piezoresistive pressure sensors exhibit non-linearity errors up to 1.5% FSR [71], and magneto-resistive sensors can reach 5.8% FSR error [17]. Lossy capacitive sensors [20] show errors up to 6% for $G_x = 1 \mu\text{S}$. Environmental factors, temperature fluctuations, and aging also degrade accuracy. For example, NTC thermistor measurements in 32 nm CMOS nodes can exhibit temperature errors up to 30 °C after one month [100].

5.2. DIC with RL Components Discussion

DICs with RL components represent an emerging field: only 1% of studies analyze errors in mutual inductive sensors [99], while 5% focus on *RL*-based advancements—interfacing inductive sensors [21], differential inductive sensors [22,98], and resolution enhancement [74,97]. The accuracy of *RL* DICs is influenced by circuit design and component properties. The key limitations observed in Section 4 are as follows:

- **Systematic errors.** Quantization errors dominate at low inductance values (1 mH–10 mH), leading to random non-linearity errors (NLE) of up to 0.70% [21]. In [99], the quantization error is inversely proportional to the reference-oscillator frequency, varying from 62.5 ns to 62.5 ns for a 16 MHz clock. These errors decrease effective resolution and introduce variability in measurements.
- **Trade-offs.** For target applications, performance factors such as accuracy, resolution, measurement time, and power consumption must be balanced. A higher resolution often requires increased clock frequency, which reduces quantization error. For example, Ref. [21] achieved 10-bit resolution with NLE < 0.3% FSS; further improvement to 10.5 bits necessitated averaging multiple measurements, which lengthened measurement time. Higher performance also typically leads to increased power consumption; in [97], raising the inductor current improved resolution but also increased power draw. Achieving linearity over wide measurement ranges is challenging due to parasitic effects. In [98], NLE was 4% FSS over the full range but <1% FSS within ± 25 mm, indicating that broader ranges degrade linearity while narrower ranges enable higher accuracy. Noise reduction often requires longer measurement times or additional filtering; averaging ten measurements in [98] reduced standard deviation from 0.011 to 0.003 but increased acquisition time, highlighting the trade-off between precision and speed.
- **Current sourcing and sinking limitations.** The MCU pin drive capacity restricts the minimum series resistance and thus impacts resolution. In [97], the ATmega328P's maximum current capacity limits the minimum series resistance to 125 Ω , which affects the *RL* time constant and restricts design flexibility and resolution, especially for low-inductance sensors.
- **Parasitic components.** Parasitic resistances in inductors and digital processor pins introduce nonlinearities that degrade accuracy, especially in the higher inductance ranges of 10 mH to 100 mH, thereby increasing overall uncertainty.

5.3. DIC with Capacitive Charge Transfer Discussion

DICs using capacitive charge transfer are discussed in two areas: operating-principles analysis (2% of works) [53,72] and recent advances (3%), including interfacing simple capacitive sensors [51], differential capacitive sensors [77], and resistive sensors [78]. Primary accuracy limitations and trade-offs identified in Section 4 include the following:

- **Parasitic capacitances.** Sensitivity to parasitic capacitances significantly increases measurement uncertainty, especially at the pF level. Parasitics depend on PCB layout and wiring between the sensor and the processor. In [53], internal calibration reduced these effects but still showed deviations of up to 0.08 FSR for 2–10 pF. In [77], residual stray capacitance caused a $\pm 4\%$ deviation for 100 pF sensors and a $\pm 0.6\%$ deviation for 1 nF sensors.
- **Temperature and supply-voltage dependence.** Accuracy is highly sensitive to temperature and supply-voltage variations. In [53], external calibration maintained deviations below 0.02 FSR under constant temperature and voltage. Internal calibration was more robust against voltage changes but still exhibited deviations

<0.01 FSR over specific ranges, demonstrating the challenge of consistent accuracy in fluctuating environments.

- Calibration techniques. Calibration reduces accuracy limitations. A two-point calibration [78] reduces non-linearity and decouples response from MCU parameters and temperature. However, systematic deviations persist for small capacitive sensors. In [77], systematic deviations were larger for smaller sensors, indicating that calibration alone cannot eliminate all errors.
- Measurement time. For small capacitances, measurement time impacts accuracy. In [53], internal calibration lengthened measurement up to 1.5 s for 2 pF. This accuracy–speed trade-off recurs: faster measurements may compromise precision in noisy settings.
- External interference. Comparing charge transfer to RC-network methods, Ref. [72] found that RC is more susceptible to capacitive interference, increasing reading dispersion. While charge transfer is more immune, careful design is required to mitigate parasitic capacitance and internal charge injection.
- Resolution. Achieving high resolution for small capacitances requires large resistances [51], which increases susceptibility to noise and interference. While charge transfer reduces noise susceptibility, maintaining high resolution over a broad range of capacitances remains challenging.

5.4. DIC Applications Discussion

DICs are increasingly recognized as viable solutions for industrial automation, health-care, and environmental monitoring applications, as summarized previously in Table 16. In recent years, emerging sensor interfaces related to health monitoring implementations have become evident. Figure 24 provides a general summary of DIC applications: health monitoring interfaces [65,83,85,87,89] account for 26.3%, followed by liquid level sensing interfaces at 15.8% [52,73,88], using USTI integrated circuits for capacitive and resistive sensors at 15.8% [102–104], gas monitoring interfaces at 10.5% [76,101], humidity sensing interfaces at 10.5% [18,75], vehicular safety systems at 10.5% [82,83], tactile sensing interfaces for industrial robotics at 5.3% [84], and distance measurement interfaces at 5.3% [105].

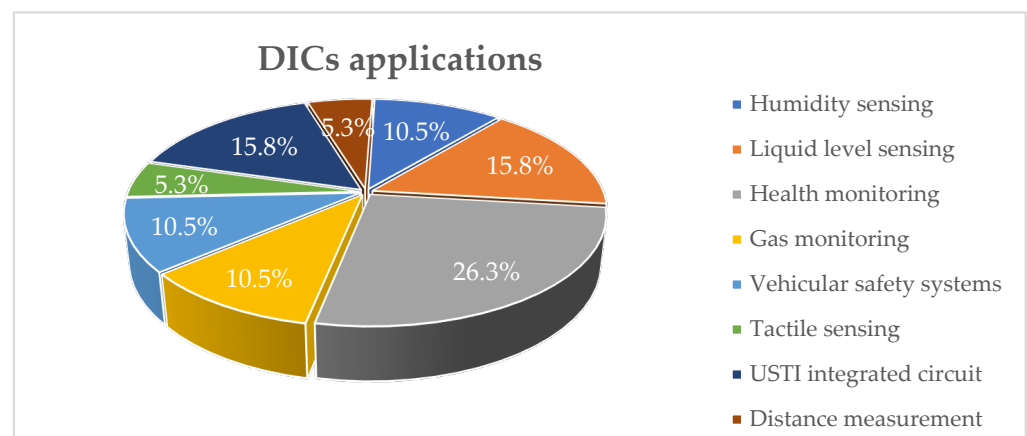


Figure 24. Applications of DICs by percentage areas.

Future challenges for DIC applications include balancing accuracy, resolution, and low power consumption, particularly in portable and wearable devices, and scaling DICs for complex systems, such as electronic skin and IoT devices, without compromising speed or reliability. As in previous sections, addressing sensor trade-offs in dynamic environments is critical, as is integrating advanced signal processing for real-time operation. Enhancing robustness for harsh conditions, such as extreme temperatures or humidity, will expand

their use in healthcare, agriculture, and industry. Overcoming these challenges demands innovative designs and interdisciplinary collaboration to maximize DICs' potential in next-generation applications.

5.5. General Discussion

Despite significant progress, key challenges remain. Table 18 organizes recent advancements and outstanding issues by DIC component. For RC-based interfaces using MCUs and FPGAs, integration has yielded low-cost, compact, and energy-efficient designs. However, improving accuracy, resolution, and reducing measurement time are ongoing concerns. As shown in Table 13, RC-based DICs exhibit measurement times from microseconds to milliseconds, limiting their suitability for time-critical applications. Quantization and non-linearity introduce relative errors between <0.01% and 5.5%. These observations highlight the need for advanced, sensor-specific calibration techniques that enhance accuracy while maintaining energy efficiency.

Table 18. A summary of critical key points observed in the review.

Category	Key Elements	Advances	Challenges	Perspectives and Recommendations
RC circuits	Resistors, capacitors, MCU, or FPGA	Low cost, compact design, energy-efficient, and simple implementation.	Enhancing accuracy and resolution, reducing measurement time, and minimizing quantization errors.	Investigate advanced calibration strategies and optimize power consumption.
RL circuits	Inductors, resistors, and MCU	Energy efficiency and accurate inductance measurements.	Addressing issues related to noise sensitivity and prolonged measurement time.	Develop strategies for noise immunity, faster measurement processes, and optimized circuit topologies.
Capacitive charge transfer	Capacitors, resistors, and MCU	Compact and efficient measurement, robust performance under controlled conditions.	Sensitivity to parasitic capacitances and systematic measurement errors.	Apply enhanced multi-point calibration methods to increase reliability and reduce systematic errors.
Digital processors	MCU or FPGA	Flexible integration, broad commercial availability, and adaptability for various applications.	Resolution variability is dependent on internal timers and power management challenges.	Evaluate trade-offs between energy consumption, performance and adjust operating frequencies according to the application.
Calibration techniques	Single-point, two-point, and three-point calibration	Improved accuracy, compensating for systematic measurement errors and enhancing reliability.	Increased complexity and higher power consumption with advanced techniques.	Explore hybrid and adaptive calibration methods tailored to sensor characteristics and application environments.
Differential and bridge-type sensors	Resistive, capacitive, and inductive differential sensors	Improved precision through reduced common-mode noise and increased reliability.	Higher complexity and cost in design and implementation.	Develop innovative DICs with compensation methods to mitigate environmental effects and parasitic influences.

For RL circuits, where inductive sensors are primarily interfaced with MCU, challenges include achieving high accuracy and resolution and reducing lengthy acquisition times. Table 14 shows that while resolutions of 10.5 to 12 bits are achievable, some systems require up to 64 measurement cycles. Although non-linearity errors are typically below 1%, the extended processing time can be a limitation. Furthermore, the integration of

active components like MOSFETs, while improving performance, adds complexity and cost. Therefore, future strategies should aim to optimize noise immunity, reduce measurement latency, and simplify circuit designs for broader applicability.

Capacitive charge transfer circuits provide an efficient interface using minimal components, making them particularly well-suited for applications that require compact and low-power designs. However, as shown in Table 15, sensitivity to parasitic capacitance remains a significant challenge, often introducing systematic errors of up to $\pm 5\%$ FSR. Nevertheless, this technique offers notable advantages, including the capability to detect small capacitance changes with resolutions as low as 0.1%. High-resolution measurements and robustness to interference are key benefits, although further research into improved calibration techniques could reduce errors and enhance reliability.

Regarding digital processors, MCUs and FPGAs are fundamental elements in DIC implementations. While they offer versatility and compact integration, challenges remain concerning timer resolution and time-to-digital conversion (TDC) precision. Performance is directly tied to processor frequency, and inadequate resolution affects measurement fidelity. Energy management strategies—shifting between idle, sleep, and active modes—are critical for balancing power efficiency and responsiveness. Research should continue to explore these trade-offs to develop optimized DIC architectures. Table 19 highlights the key MCU characteristics necessary to enhance the performance of DICs in terms of accuracy, precision, resolution, and energy efficiency. These features are essential for ensuring reliable signal acquisition while maintaining low power consumption in embedded sensing applications.

Table 19. Recommended MCU features for optimizing time-to-digital conversion in DICs [6,37].

Component	Required Features
General-Purpose Digital I/O Pins	Programmable in three states: (1) Output at logic level “0” (LOW). (2) Output at logic level “1” (HIGH). (3) Input with high-impedance state (“HZ”).
Digital Pin P_{d1}	- Must include a Schmitt trigger (ST) with defined upper and lower threshold voltages (V_{TH} and V_{TL}), enabling comparator-like behavior. - Should be associated with an external interrupt and/or a capture module. This configuration ensures a quantization error of $\pm T_{ref}$ (one timer count) during time-to-digital conversion.
Reference Oscillator	- Must be a high-frequency crystal oscillator (tens of MHz). - Should exhibit a stability coefficient greater than ± 50 ppm.
Timer Module	- Should have a minimum resolution of 16 bits. - Must be initialized and synchronized at the start of each discharge phase to prevent quantization error at the beginning of the time-to-digital conversion.
Sleep/Idle Mode	- Should allow the CPU and internal peripherals to be disabled, except for the timer. - Must be employed during the time-to-digital conversion to reduce noise effects caused by CPU and peripheral activity.

Table 19. *Cont.*

Component	Required Features
CPU Core	• Must offer configurable options for selecting and scaling internal or external oscillator sources. • Must support assigning the high-frequency external reference oscillator to the timer. Higher frequency improves resolution in time-to-digital conversion but increases power consumption.
Power Supply	• Must be stable and preferably derived from a low-dropout (LDO), low-noise, low-power linear regulator. • A decoupling capacitor of the value the manufacturer recommends should be placed at the digital processor power supply terminals.

Regarding cost, DICs inherently support low-cost implementations by minimizing the required components. This advantage is particularly beneficial in IoT applications, where battery-powered devices must operate over long durations with limited energy resources. However, DICs remain susceptible to external interferences and power supply fluctuations, which can compromise measurement integrity. Addressing these issues through improved shielding, filtering, and calibration remains a key area of focus.

Despite recent advances, several research gaps still limit the development of high-performance DICs. Significant challenges include

- Systematic errors caused by instrument drift, environmental variation, and sensor aging necessitate adaptive real-time compensation.
- Quantization and trigger noise that limit the accuracy of time-based measurements, particularly for high-resistance and low-capacitance sensors.
- Parasitic impedance, including stray capacitance, lead resistance, and mutual coupling, reduces performance in remote and miniaturized sensors, underscoring the necessity for simplified, parasitic-tolerant designs.
- Energy and cost constraints, and IoT- and battery-powered nodes require ultra-low-power, low-complexity implementations.
- Intrinsic non-linearities in sensors require advanced linearization strategies, such as embedded machine-learning engines.
- Accurate measurement of sub-ohm resistances and sub-picofarads capacitances remains challenging, necessitating specialized front-end architectures.
- Power-aware integration of wireless and IoT platforms imposes strict power-management overheads, necessitating holistic, energy-aware design methodologies.

Future research directions should prioritize the development of hybrid and adaptive calibration methods tailored to specific sensor characteristics. Software-based error compensation techniques for time-to-digital conversion can also enhance measurement accuracy under varying conditions without increasing hardware complexity. Moreover, integrating newer technologies, such as precise comparators or low-resistance MOSFETs, may improve performance while preserving the simplicity of DICs. Addressing these challenges is essential for the next generation of efficient, accurate, and versatile DIC-based sensor systems.

6. Conclusions

Direct interface circuits (DICs) offer a compact, cost-effective, and power-efficient solution for measuring resistive, capacitive, and inductive sensors by directly interfacing these elements with digital processors such as MCUs or FPGAs. Key improvements

include enhanced accuracy through various calibration techniques reported in this review. However, challenges persist with resolution and uncertainty due to interference, noise, temperature fluctuations, and the complexity of sensor integration. Future opportunities lie in developing hybrid calibration methods, optimizing energy-efficient hardware, and reducing noise in time-to-digital conversion. Compared to traditional conditioning schemes, DICs require fewer components and exhibit lower energy consumption, making them highly attractive for battery-powered or resource-constrained applications in the Internet of Things (IoT), industrial automation, and medical devices. In conclusion, the effective integration of sensors through DICs continues to be a highly active research area, with potential improvements in calibration methodologies, low-power topologies, and advanced noise compensation methods.

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